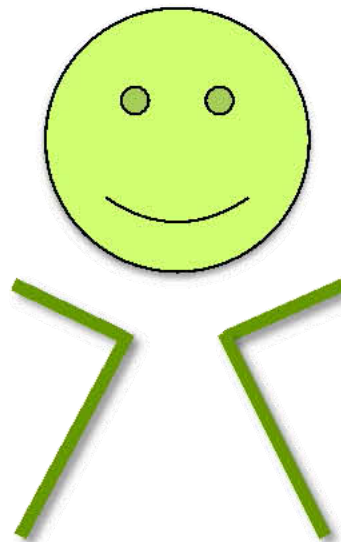




Human-Computer Interaction

an overview



“Design, v.: What you regret not doing later on.”

/usr/games/fortune

How about the means of interaction
between user(s) and software?



human-computer interaction

Human

a person who tries to accomplish a goal

human-computer interaction

Computer

runs applications (software)

human-computer interaction

Computer

runs applications (software)

locally *versus* remotely



first game developed on PDP (1960s)



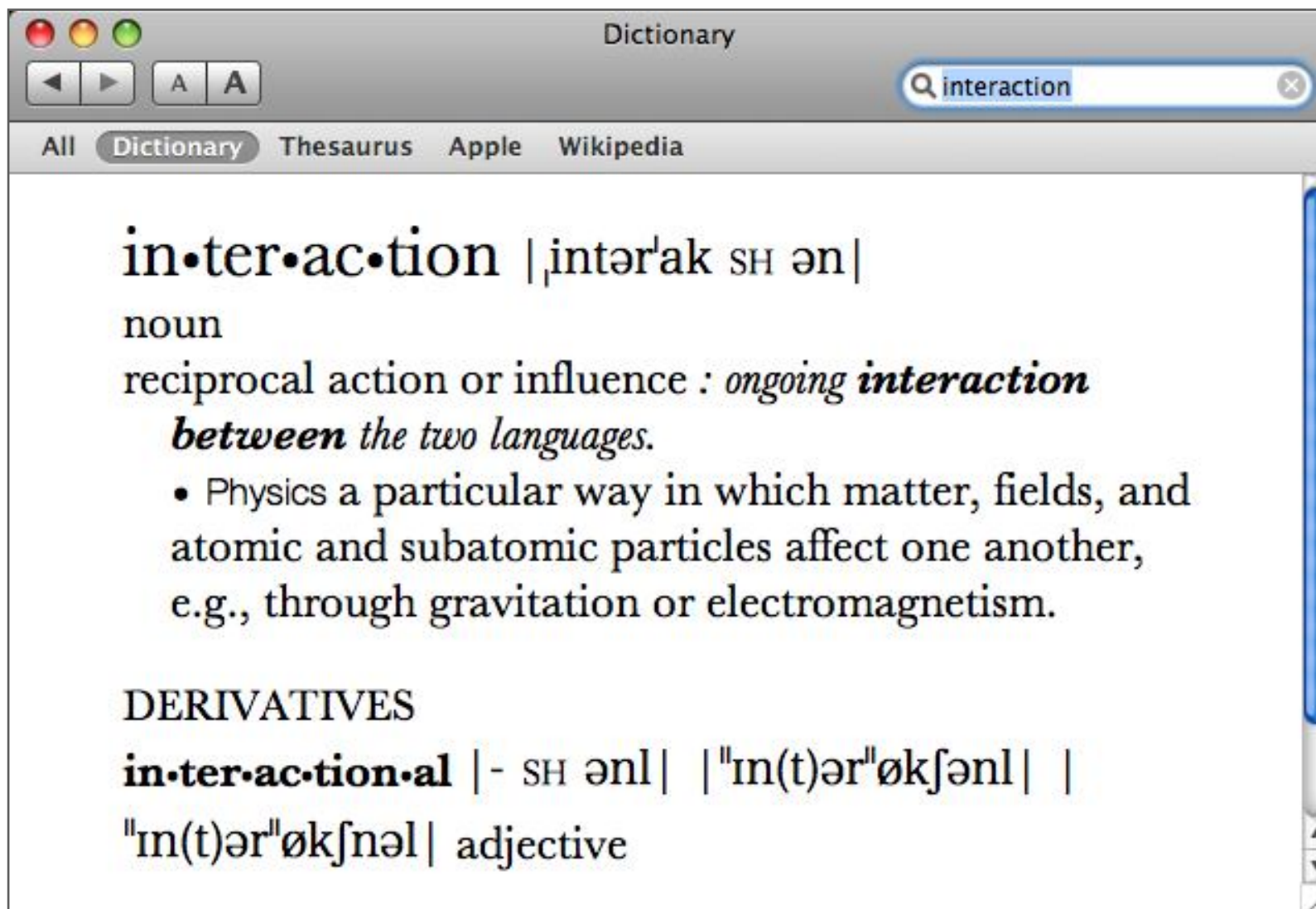
Master on Software Engineering :: Human-Computer Interaction



human-computer interaction

Interaction

“dialogue” between humans and computers



human-computer interaction

The interaction between user(s) and application(s) is achieved via an interface – **user interface**

human-computer interaction

The interaction between user(s) and application(s) is achieved via an interface – **user interface**

API (Application Programming Interface)

versus

UI (User Interface)

UI (User Interface)

human-computer interaction

Types of user interfaces:

conventional (classical) UI – *e.g.*, desktop



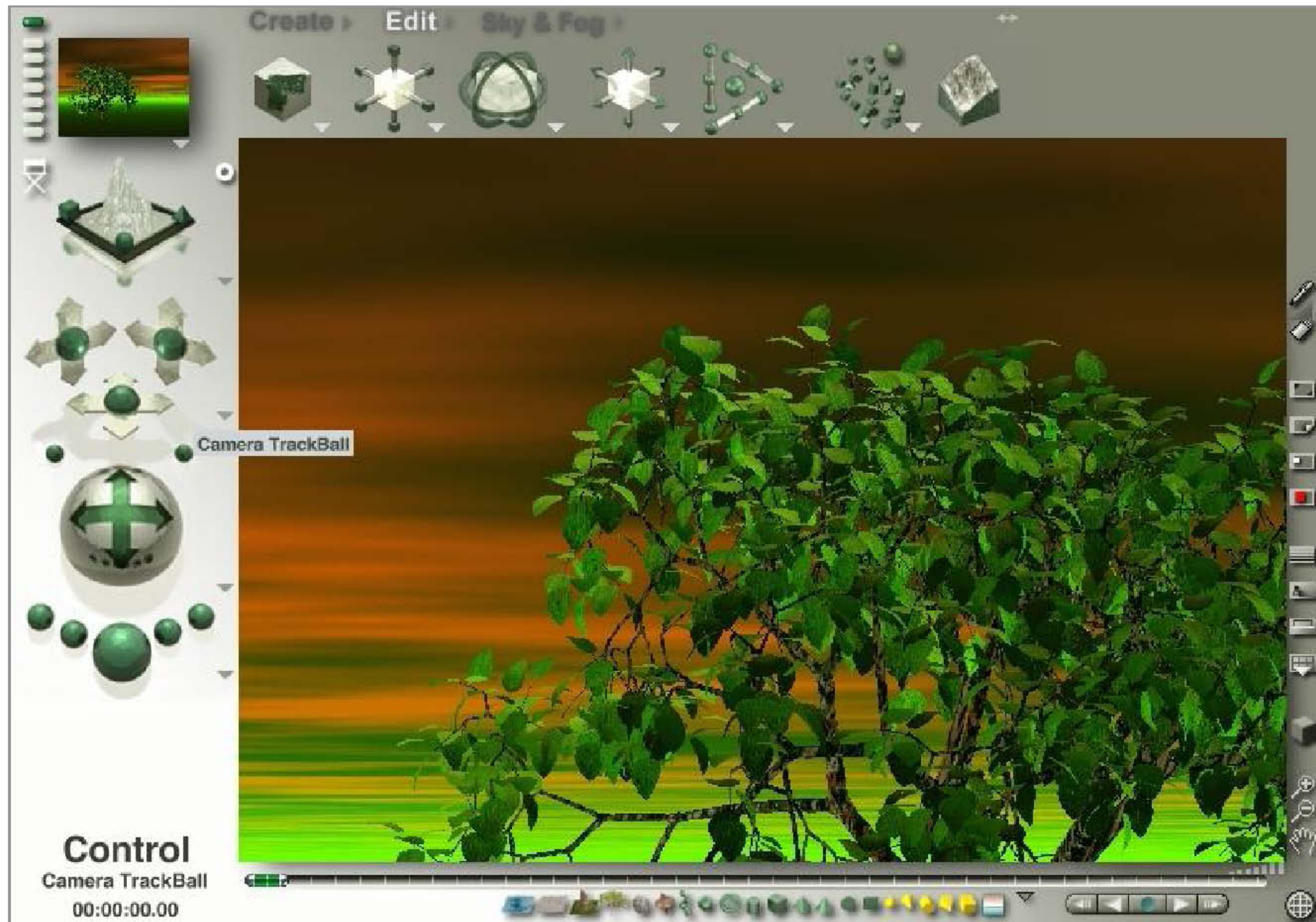
Graficon (Douglas Engelbart, 1963)

human-computer interaction

Types of user interfaces:

conventional (classical) UI – *e.g.*, desktop

WIMP (Window Icon Menu Pointer) paradigm,
interaction via keyboard & mouse + additional devices,
graphical representations (bitmap vs. vectorial),
support provided by the OS kernel vs. a desktop system,
multi-tasking

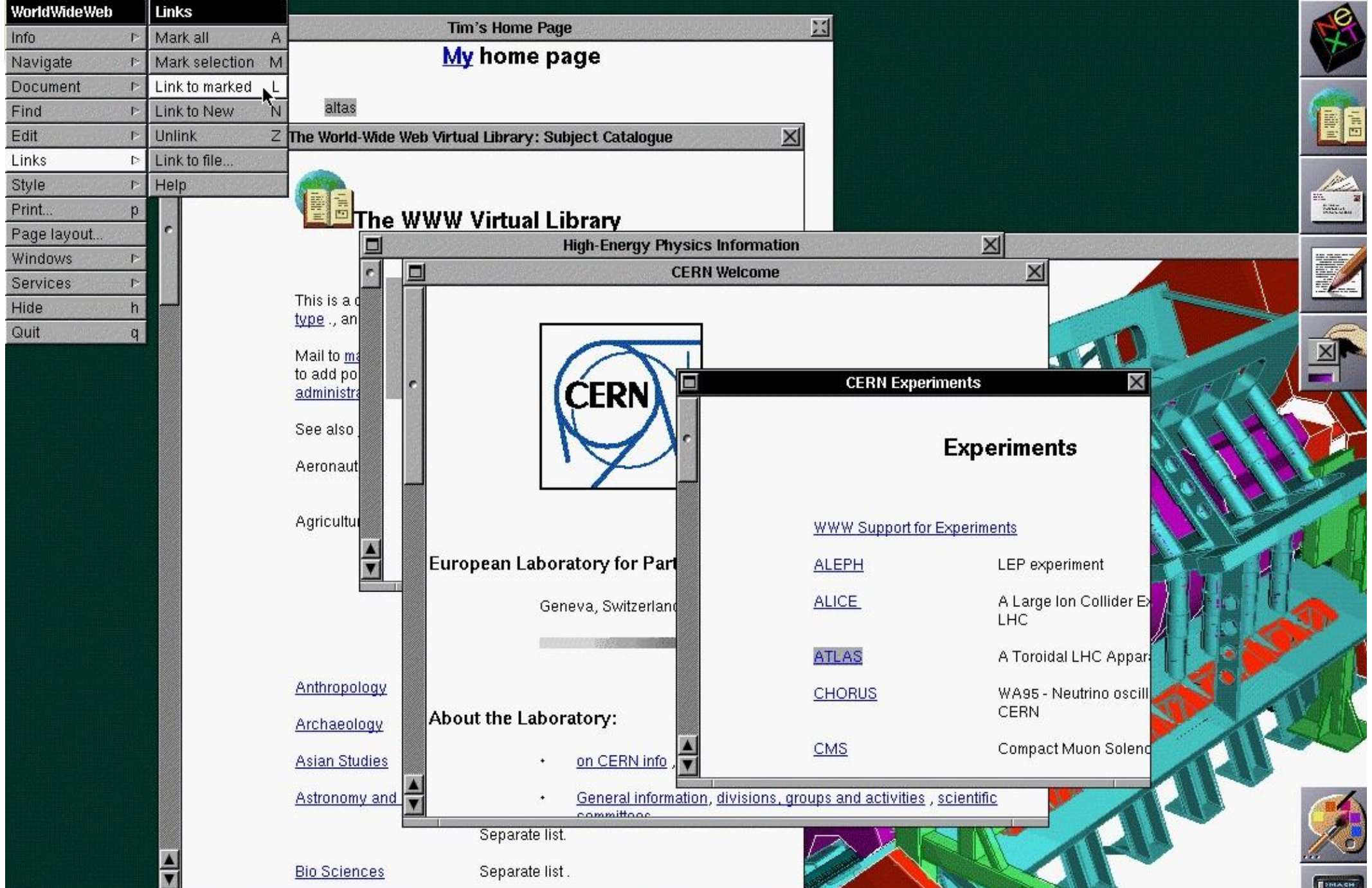


Bryce (Kai Krause, 1997)

human-computer interaction

Types of user interfaces:

Web interface



WorldWideWeb system (Tim Berners-Lee, ~1990)

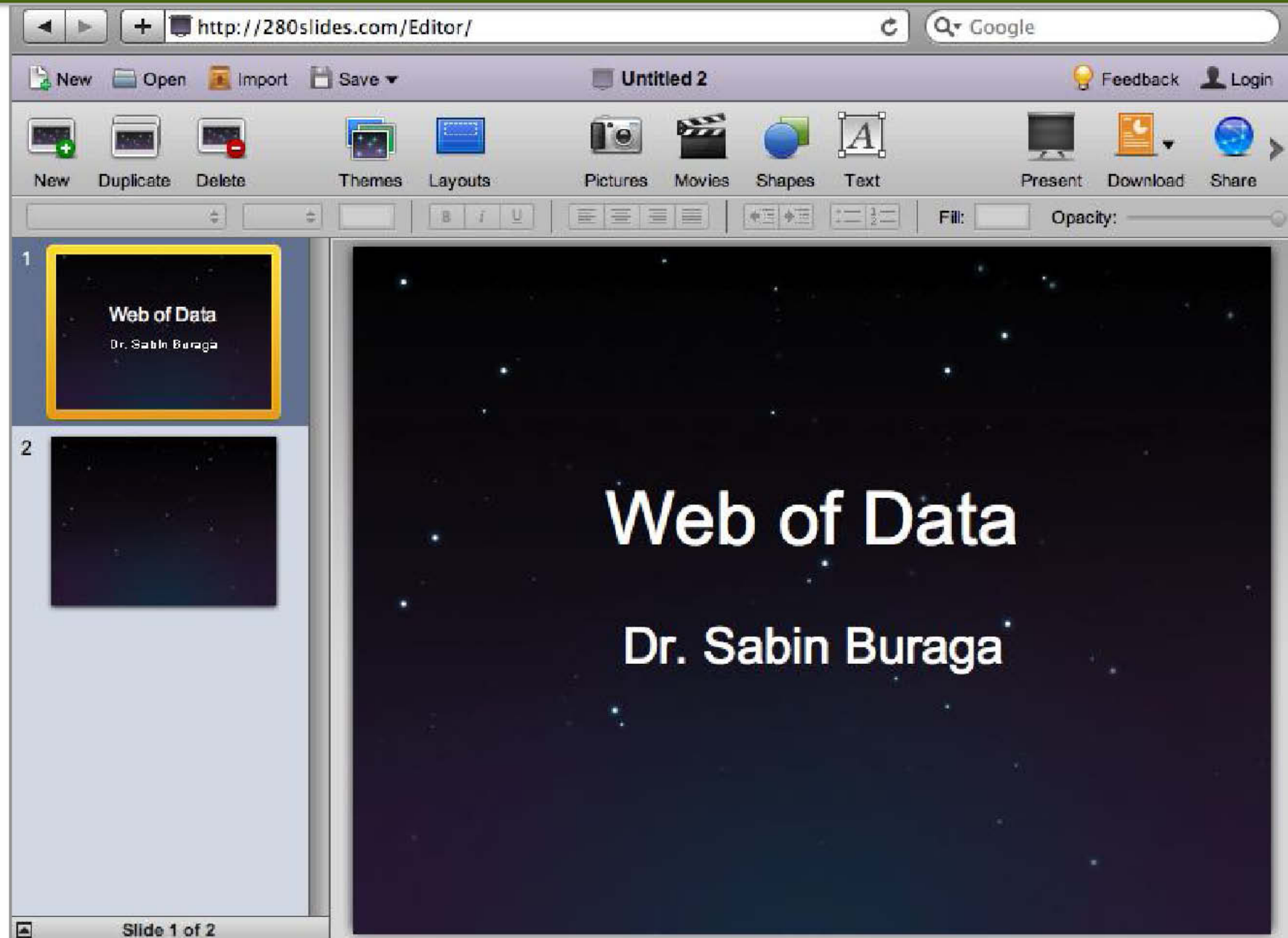
www.w3.org/History/1989/proposal.html

human-computer interaction

Types of user interfaces:

Web interface

browser – limited controls provided by (X)HTML,
hypertext/hypermedia,
RIA (Rich Internet Applications),
(a)synchronous interaction,
availability – world-wide audience,
standards



280slides (2010)

human-computer interaction

Types of user interfaces:

mobile user interface

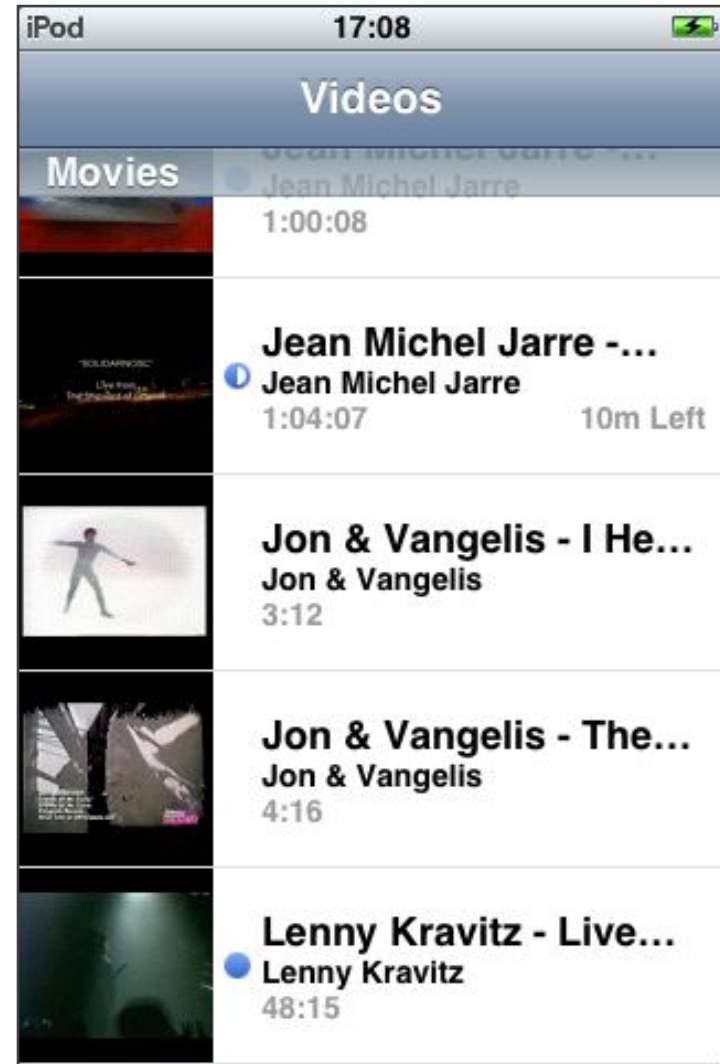
human-computer interaction

Types of user interfaces:

mobile user interface

limited resources/features:
resolution, memory, I/O, power etc.,
one task vs. multi-tasking,
off-line vs. on-line,
task-oriented software

human-computer interaction



human-computer interaction

Types of user interfaces:

unconventional UI

human-computer interaction

Types of user interfaces:

unconventional UI

new interaction paradigms + new expectations,
haptic, gesture, locomotion, auditory & voice, tangible UI,
augmented and virtual reality,
physical computing,
ambient intelligence (AmI)

Tracking Speed



Double-Click Speed



Scrolling Speed



One Finger

- ☒ Tap to Click
- ☒ Dragging
- ☒ Drag Lock
- ☒ Secondary Click Bottom Left Corner ▾

Two Fingers

Scroll

Rotate

Pinch Open & Close

Screen Zoom Options...

- ☒ Secondary Tap

Three Fingers

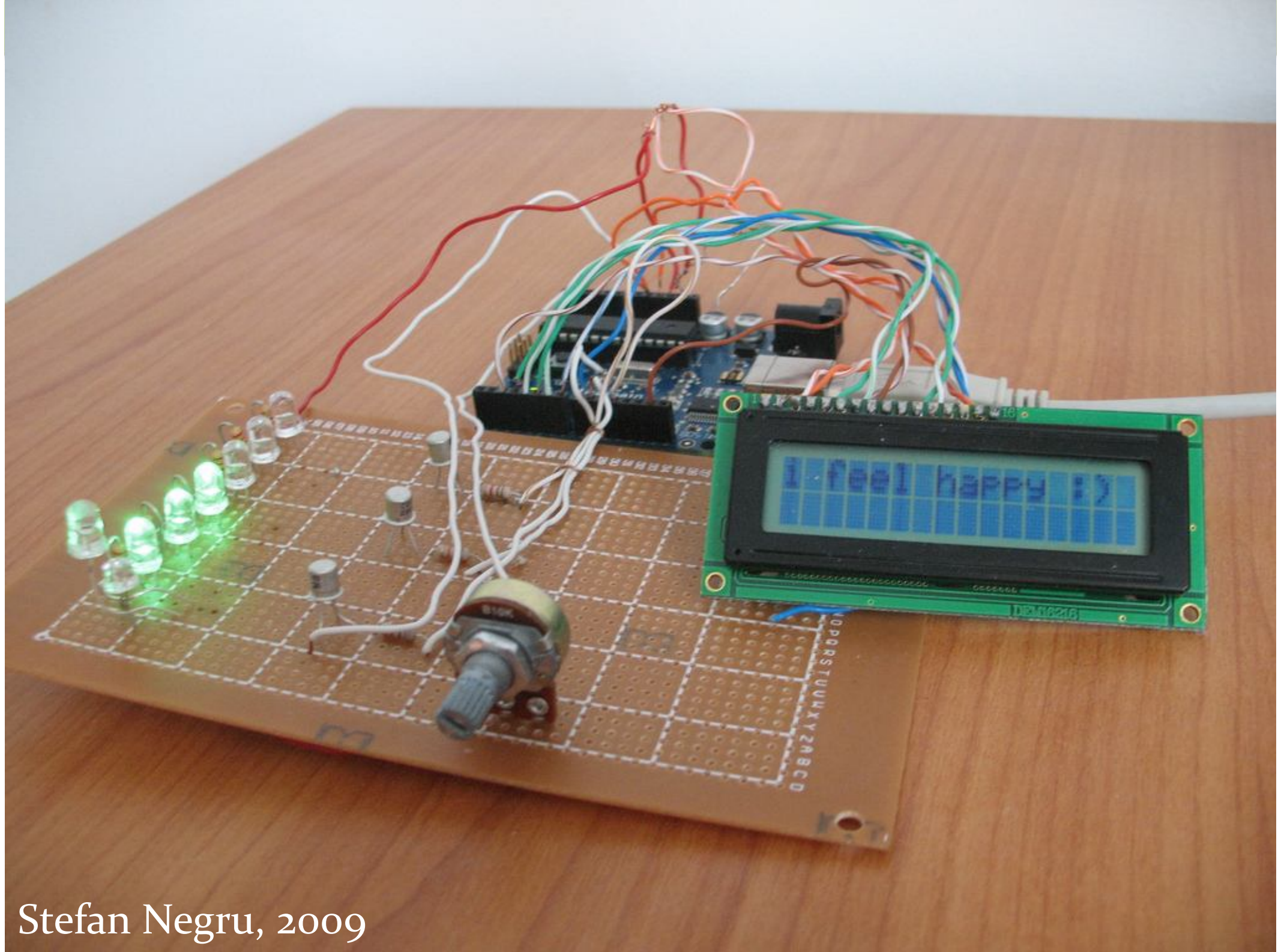
Swipe to Navigate

Four Fingers

Swipe Up/Down for Exposé

Swipe Left/Right to Switch Applications





Stefan Negru, 2009

human-computer interaction

User – desktop, Web, mobile,... – interface

part of a program/application

- at the desktop, Web, mobile device,... level –

which permits users to expose their intentions (goals) with respect to the actual software and to interpret the results of certain actions performed by the machine

 **Writing CD-R**
Remaining 312 of 657 MB

 **Downloading file1.zip**
Remaining 6.8 of 8.1 MB

 **Copying files**
Remaining 6 of 34 files

Connection:  online (DSL) 

Mail:  3 new

Chat: user@chat.server
online (available)
james_hetfield
disconnected

Browse the web:
URL / bookmark / search key
 

Activities / applications:

-  **Web browser**
Mozilla Firefox
-  **Chat**
Gaim
-  **E-Mail**
Evolution
-  **News reader**
Liferea

 other 

Type: ☐ text documents ☐ spreadsheets
☐ audio files ☐ movies ☐ pictures

Tags: ~~opera~~ opera pink floyd rock

Files:

-  **The Wall (the movie)**
Pink Floyd
-  **The Wall (documentary)**
Pink Floyd
-  **Comfortably Numb**
Pink Floyd
The Wall
-  **Hey You**
Pink Floyd
The Wall

human-computer interaction

User – desktop, Web, mobile,... – interface

perceived by the user not only as a visual part
of a certain software application

human-computer interaction

User – desktop, Web, mobile,... – interface

from the point of view of an user,
it represents the entire system

the application *per se*

human-computer interaction

User – desktop, Web, mobile,... – interface

useful

human-computer interaction

User – desktop, Web, mobile,... – interface

usable

human-computer interaction

User – desktop, Web, mobile,... – interface

used

human-computer interaction

Everyone who uses a (software) tool
is conducted by a **motivation**

PLACEOPEDIA

Connecting Wikipedia articles with their locations

+ Add your place Random place

Map Satellite Hybrid

Search for a location Search

Recent Additions on Map

Darabani (Wikipedia Article)
added 07:55, 27th March 2006 by Radu Sarghie Permalink Incorrect?
Husi (Wikipedia Article)
added 07:51, 27th March 2006 by Radu Sarghie Permalink Incorrect?
Pascani (Wikipedia Article)
added 07:48, 27th March 2006 by Radu Sarghie Permalink Incorrect?
Roman (Wikipedia Article)
added 07:47, 27th March 2006 by Radu Sarghie Permalink Incorrect?
Focsani (Wikipedia Article)
added 07:43, 27th March 2006 by Radu Sarghie Permalink Incorrect?
Braila (Wikipedia Article)
added 07:42, 27th March 2006 by Radu Sarghie Permalink Incorrect?

flickr



17469 photos were found on Flickr.



Explore

The top 100 twines are ranked based on the popularity of their items contributed to the twine within the last week.



1

Cool
4239 Items | 1861 Members

This is a twine for sharing unusually cool things we find on the Web. To avoid "spamming" the group, please: 1. Do not use this twine to ...

Tags [cool](#), [art](#), [design](#), [technology](#), [fashion](#),



2

Web 3.0 - Semantic Web
1965 Items | 3854 Members

This twine is about the emerging Semantic Web -- and Web 3.0, the next-evolution of the Web.

Tags [Web](#), [World Wide Web](#), [internet](#), [dataweb](#), [Web 3.0](#), [Semantic Web](#), [semantics](#), [Web 3G](#),



3

Social Networking Software and Services
2555 Items | 1504 Members

This is a twine for discussing things related to social networking software and services on the Web. Market data about this category as ...

Tags [technology](#), [social network](#), [software](#), [social](#)



4

Web Industry Trends
1412 Items | 1309 Members

This public twine is for sharing news articles or other information that are of special interest or value in understanding where the Web ...

Tags [venture capital](#), [startups](#), [silicon valley](#).

finding an entity: information, object, user, etc.

Master on Software Engineering :: Human-Computer Interaction



(self)learning/instruction



performing a process – *e.g.*, a (business) transaction



social interactions – at a real and/or virtual level

The screenshot shows the Twitter homepage with a search for "open linux". The top navigation bar includes links for Home, Profile, Find People, Settings, Help, and Sign out. The main content area features a "What are you doing?" input field with a character count of 140, an "update" button, and a "Latest" tweet from a former student. Below this, "Real-time results for open linux" are displayed, including tweets from syafthegEEK, linuxalive, openinnovation_, and Nerd6. A sidebar on the right shows the user's profile (busaco) with 74 following, 290 followers, and 2,553 tweets. It also lists navigation links (Home, @busaco, Direct Messages, Favorites), a search bar with "open linux", and saved searches for "busaco" and "semantic web". A tooltip for the "openinnovation_" tweet shows its URL.

twitter

Home Profile Find People Settings Help Sign out

What are you doing? 140

Latest: Yesterday, I had a nice meeting with a former student (Silviu Homoceanu), now PhD student in Germany. about 3 hours ago

update

Real-time results for **open linux** + Save this search

 syafthegEEK RT @linuxalive: **Open** source M&A: The scorecard to date **#linux** <http://bit.ly/9trGr>
13 minutes ago from twirl

 linuxalive **Open** source M&A: The scorecard to date **#linux** <http://bit.ly/9trGr>
34 minutes ago from API

 openinnovation_ Kandidaten für **Open** Innovation Awards gesucht – **Linux**: **Linux**-Forum <http://is.gd/2o8wS>
42 minutes ago from API

 Nerd6 2. Okt. **#Open**-Innovation-Summit in **#Paris** sucht **#Open**-Source **#Projekte** die sich vorstellen wollen. !OS !flossHH **#Linux** – <http://is.gd/2o7yl>
about 1 hour ago from Identica

busaco

74 290 2,553
following followers tweets

Trazz-ler Buzz
n. tracking popular travel destinations via Twitter.

Home

@busaco

Direct Messages 12

Favorites

open linux

Saved Searches

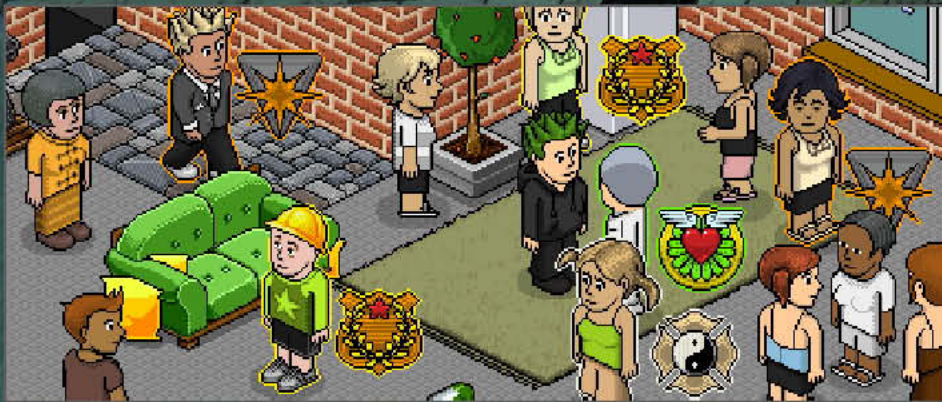
busaco

semantic web

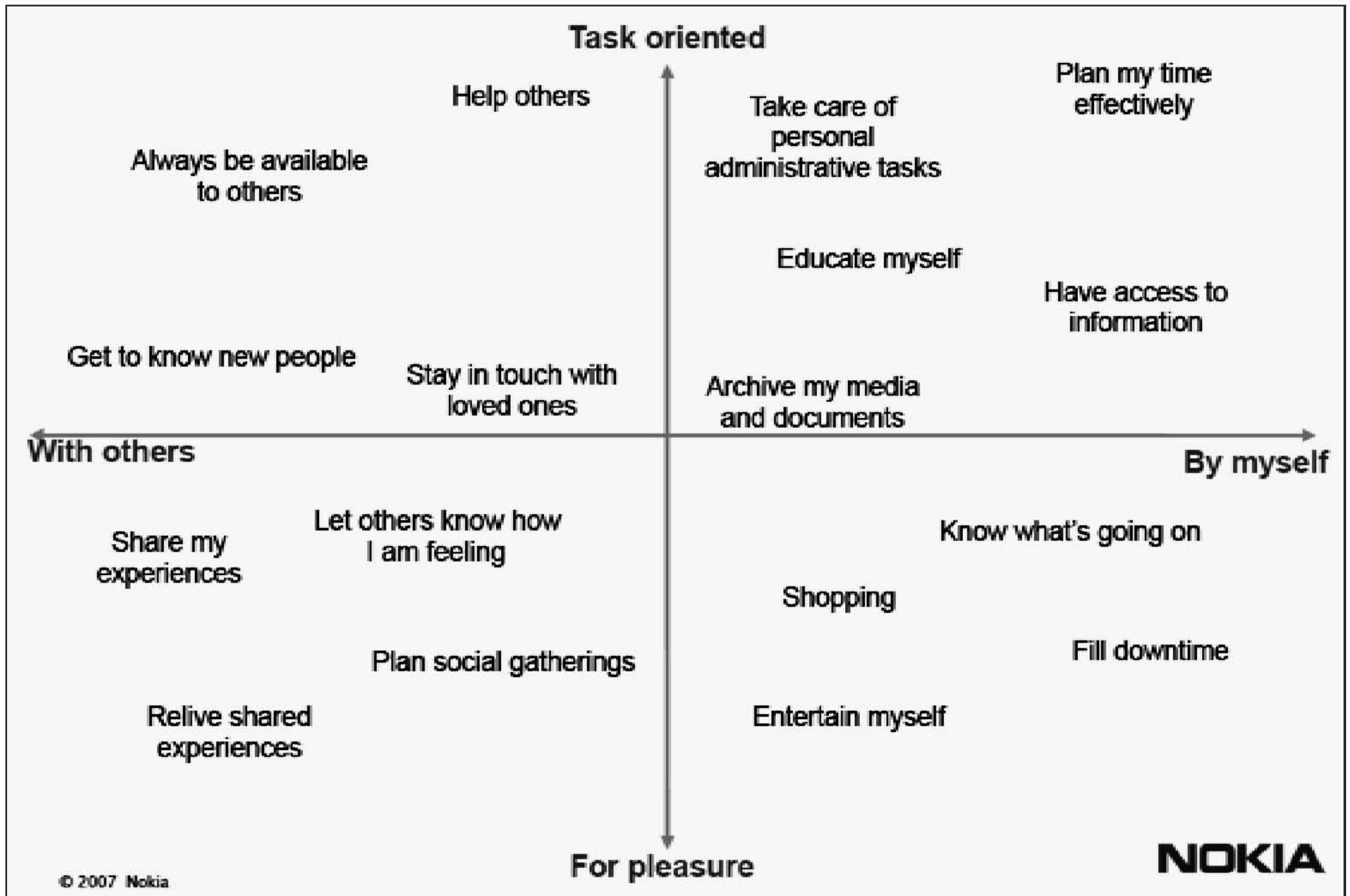
http://twitter.com/home?status=@openinnovation_%20&in_reply_to_status_id=3401906842&in_reply_to=openinnovation_

#fact

creating an artifact:
(micro)blog note, picture, article, source-code,...



entertainment – individual vs. community



UX – User Experience

Define the perception of a product/service from the point of view of person(s) who use it and the apparent pleasure/satisfaction

UX – User Experience

“Every product that is used by someone has a user experience: newspapers, ketchup bottles, reclining armchairs, cardigan sweaters.”

James Jesse Garrett, 2003

UX – User Experience

Cannot design a user experience,
only design **for** a user experience

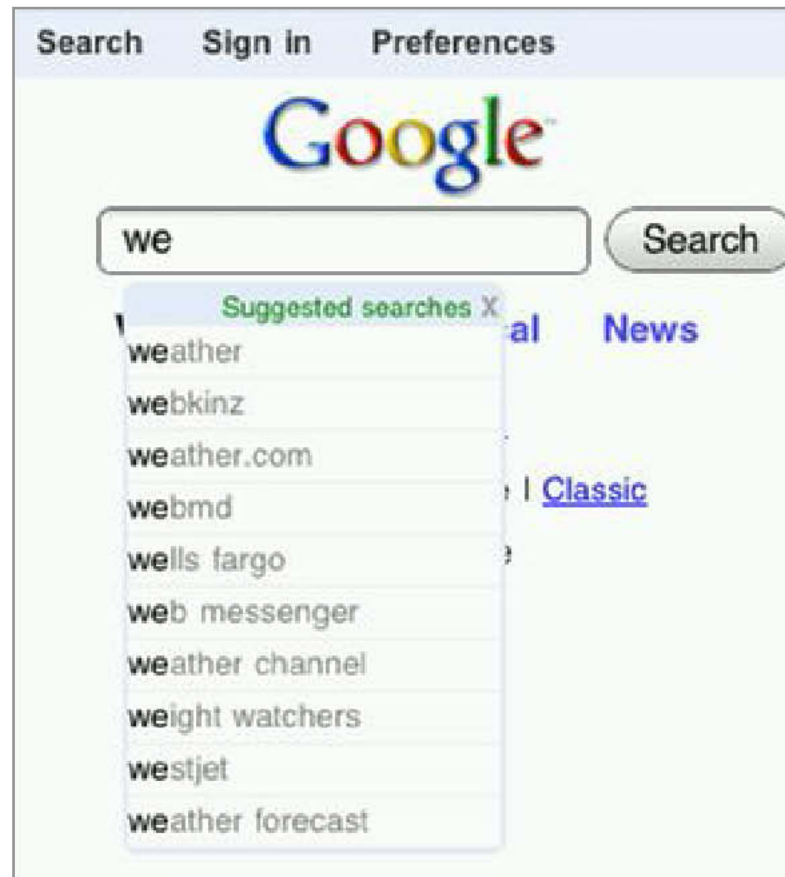
UX – User Experience

experience = **expectation** ▶ user is satisfied



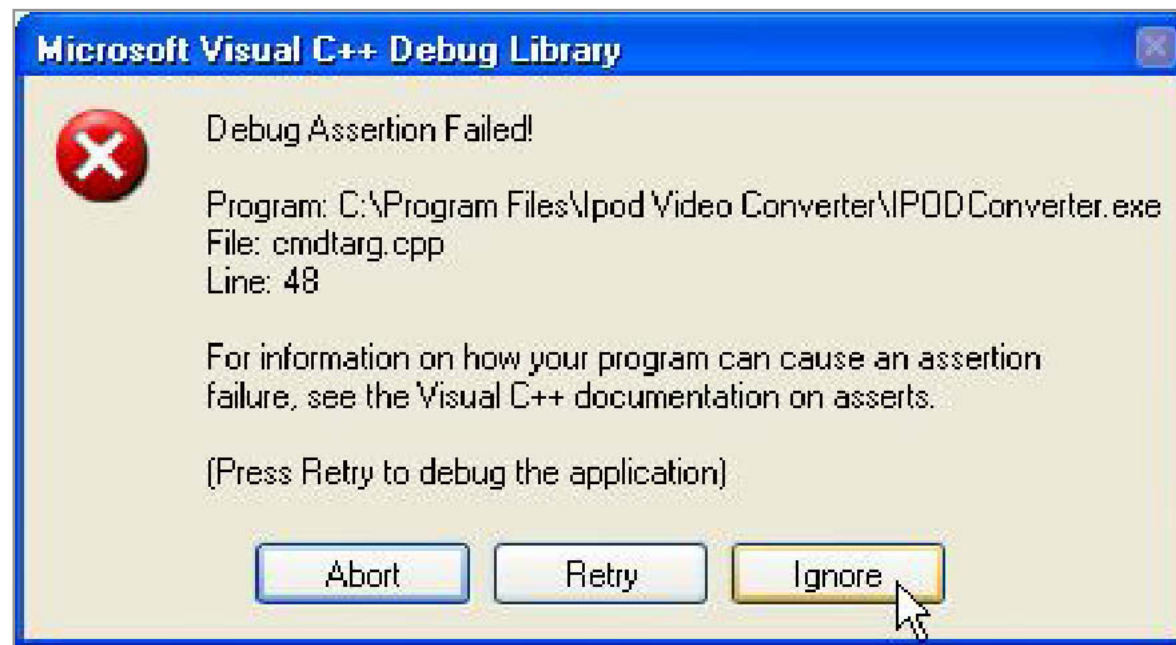
UX – User Experience

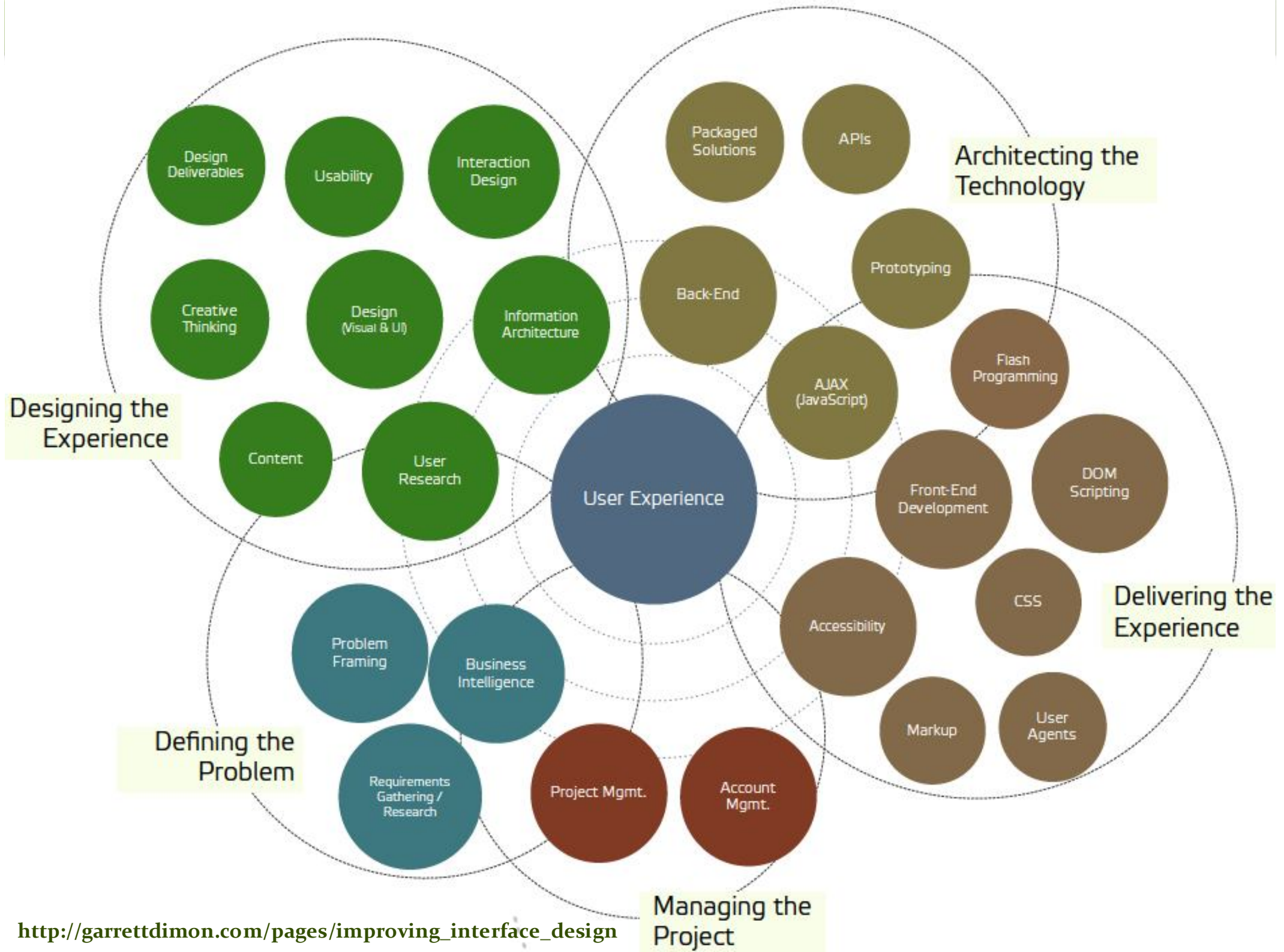
experience > expectation ► user is delighted



UX – User Experience

experience < expectation ► user is dissatisfied





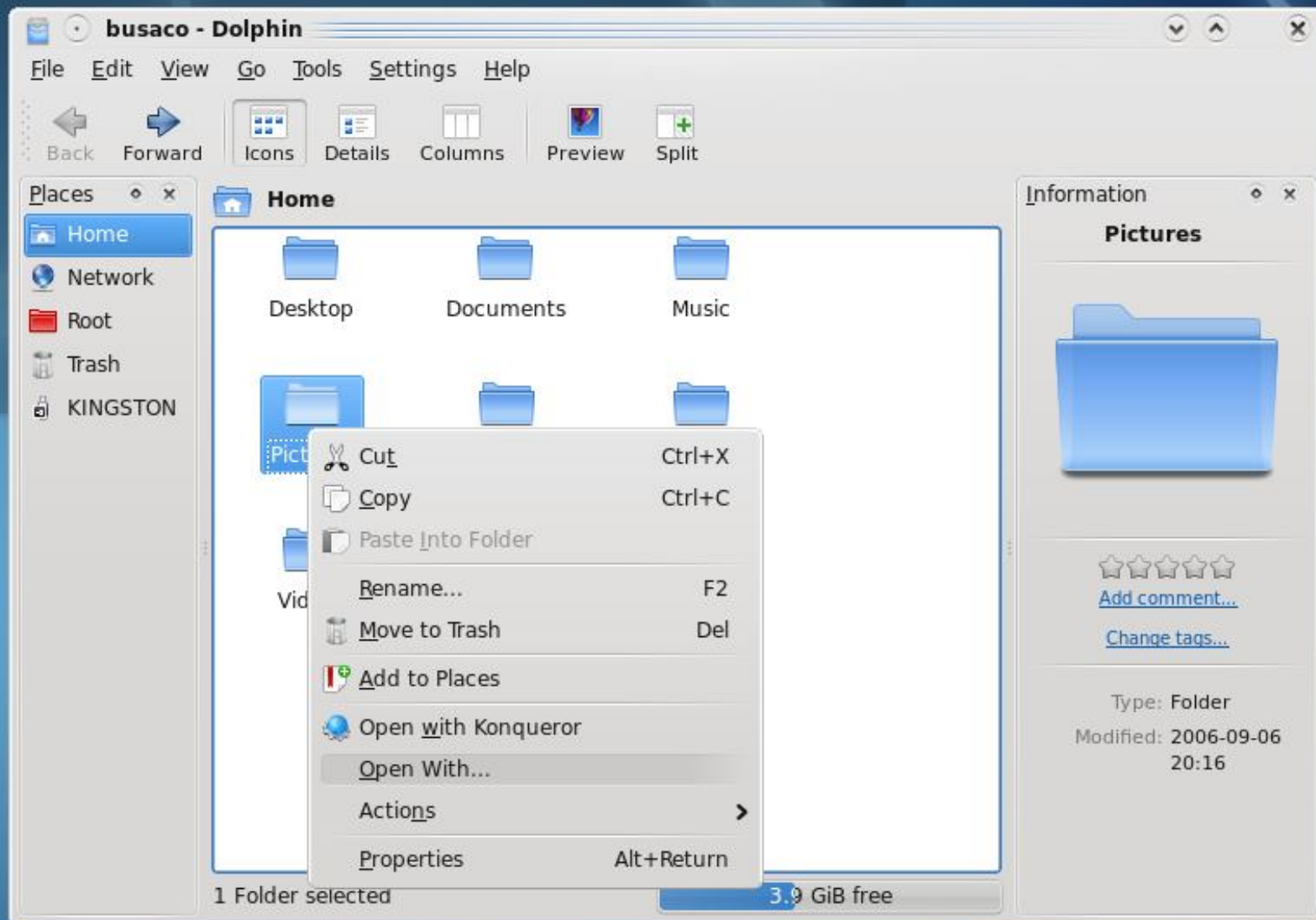
IA – Information Architecture

The necessity of organizing the information
desired by the users

IA – Information Architecture

The solutions of presenting the content regard the **IA – Information Architecture**

Desktop



new-captures - Dolphin

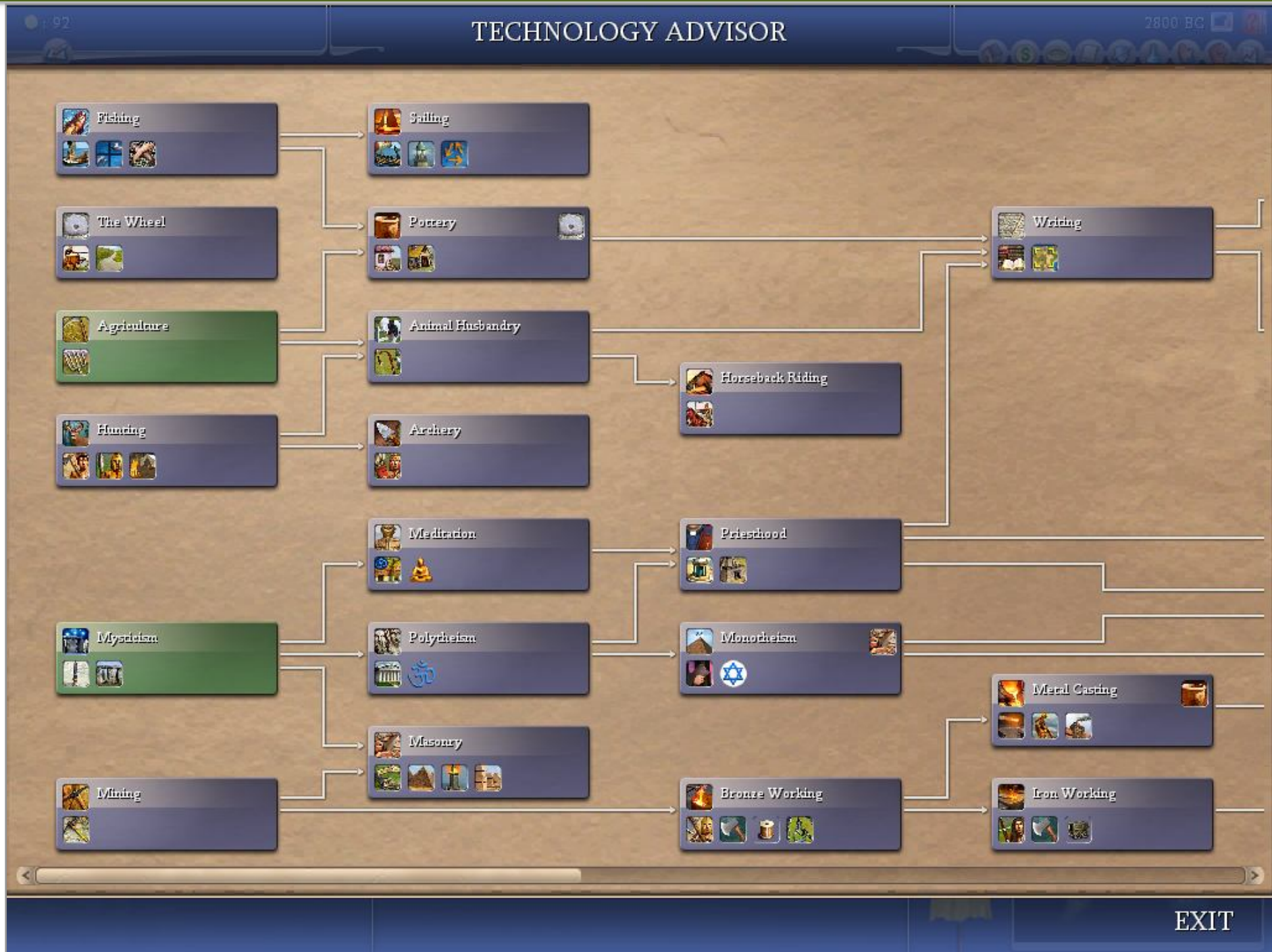


busaco - Dolphin



21:12
Wed, 6 Sep





IA – Information Architecture

We can adopt different techniques regarding
data visualization

IA – Information Architecture

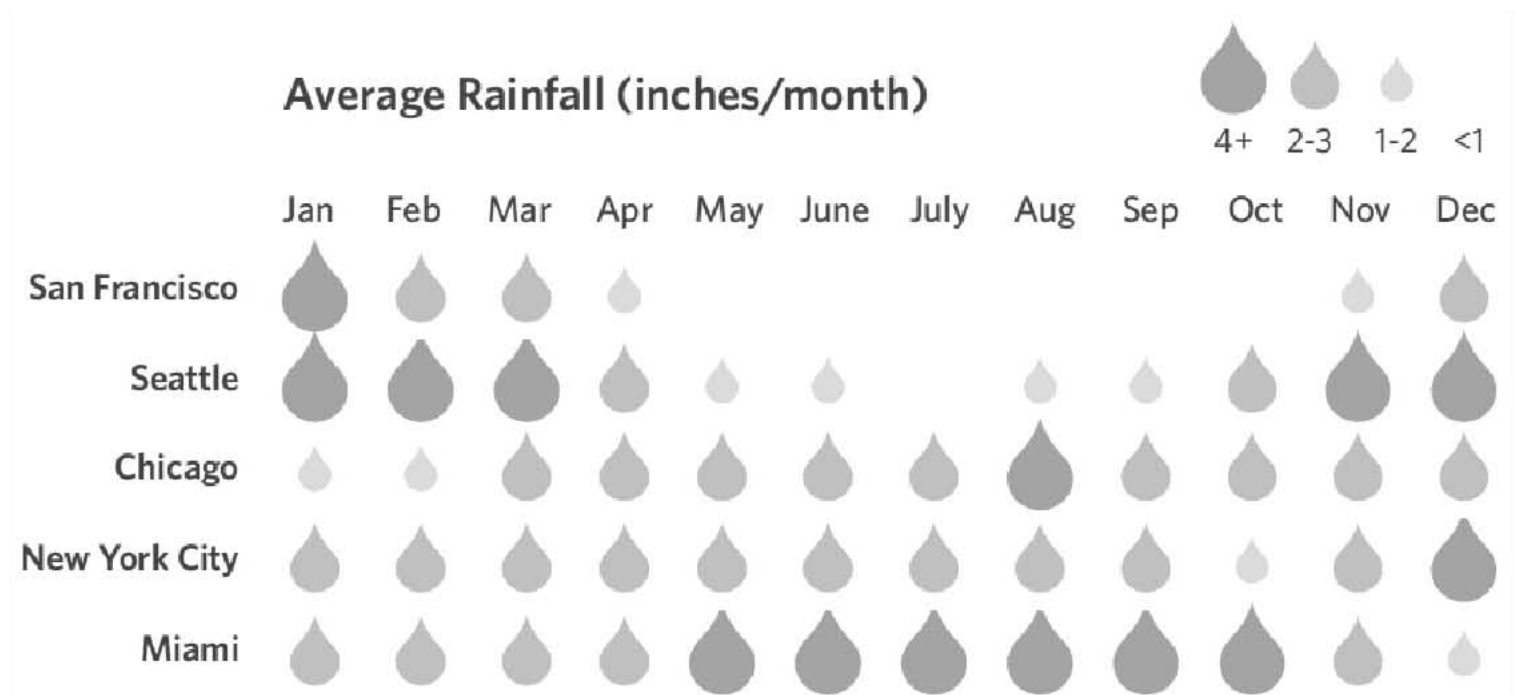
We can adopt different techniques regarding
data visualization

visual design

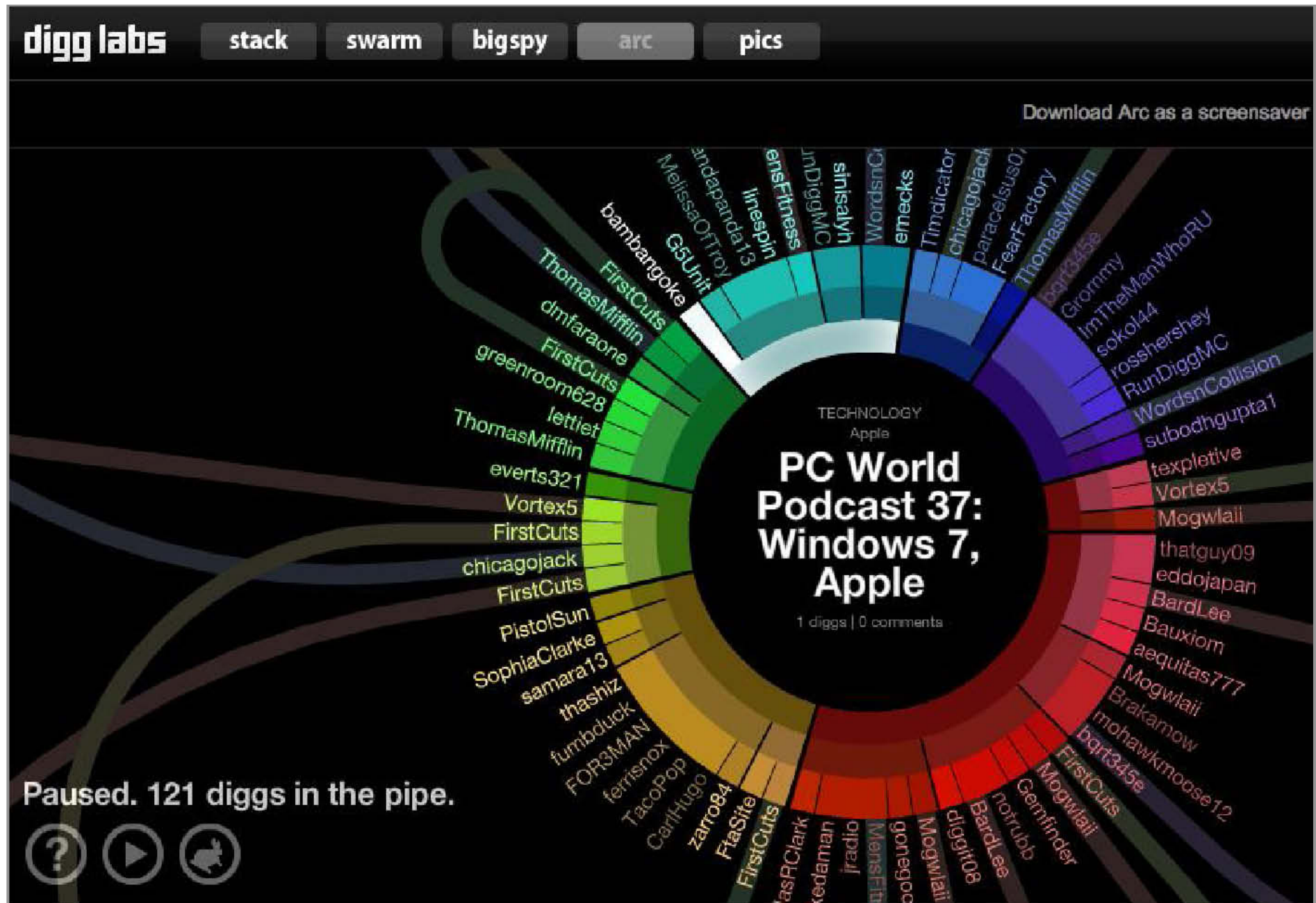
visual representation
infographics

4.35 3.17 3.06 1.37 0.19 0.11 0.03 0.05 0.20 1.22 2.86 3.09
 5.35 4.03 3.77 2.51 1.84 1.59 0.85 1.22 1.94 3.25 5.65 6.00
 1.53 1.36 2.69 3.64 3.32 3.78 3.66 4.22 3.82 2.41 2.92 2.47
 3.17 3.02 3.59 3.90 3.80 3.65 3.80 3.41 3.30 2.88 3.65 3.42
 2.01 2.08 2.39 2.85 6.21 9.33 5.70 7.58 7.63 5.64 2.66 1.83

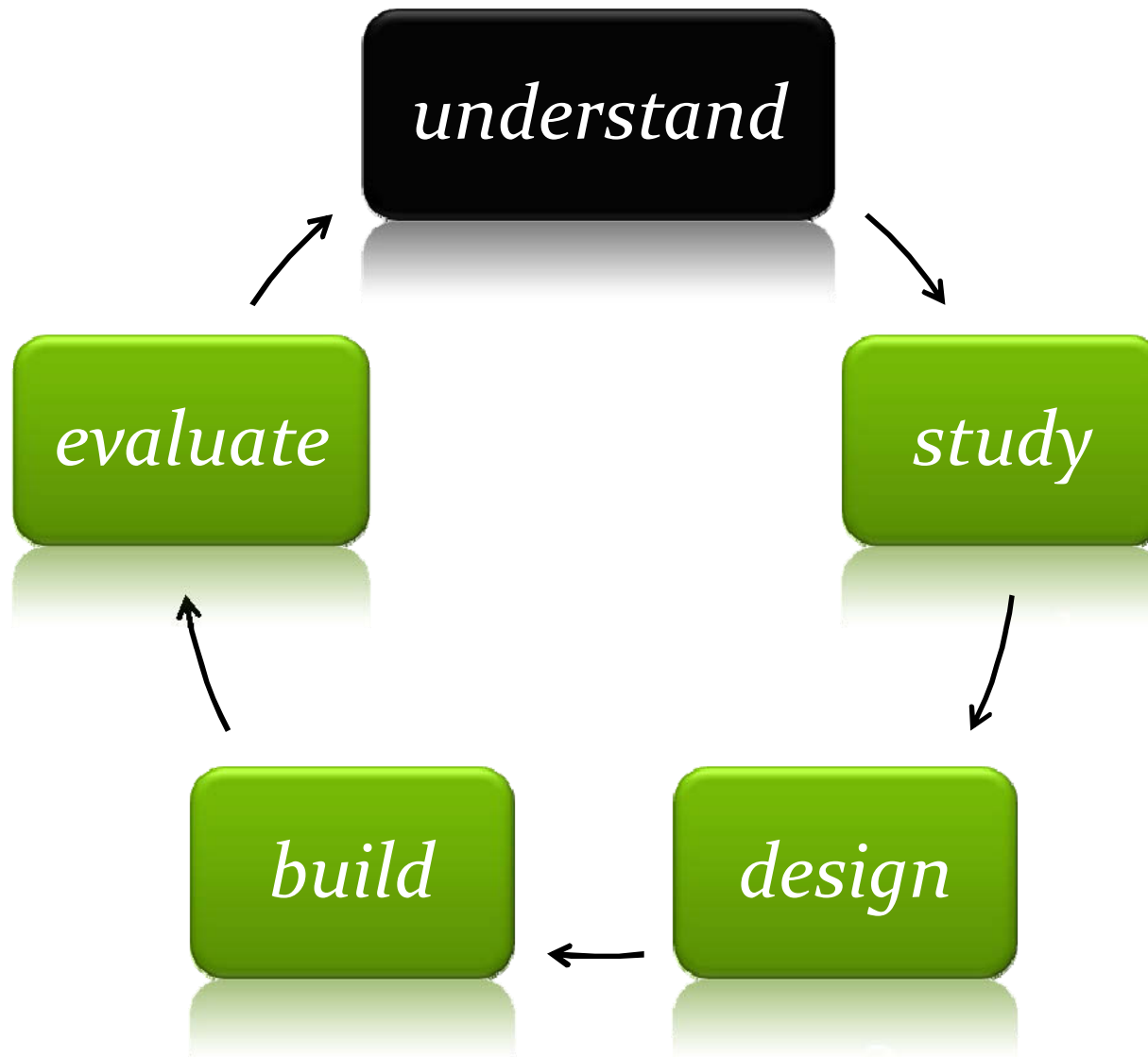
Average Rainfall (inches/month)												
	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
San Francisco	4.35	3.17	3.06	1.37	0.19	0.03	0.06	0.05	0.20	1.22	2.86	3.09
Seattle	5.35	4.03	3.77	2.51	1.84	1.59	0.85	1.22	1.94	3.25	5.65	6.00
Chicago	1.53	1.36	2.69	3.64	3.32	3.78	3.66	4.22	3.82	2.41	2.92	2.47
New York	3.17	3.02	3.59	3.90	3.80	3.65	3.80	3.41	3.30	2.88	3.65	4.42
Miami	2.01	2.08	2.39	2.85	6.21	9.33	5.70	7.58	7.63	5.64	2.66	1.83



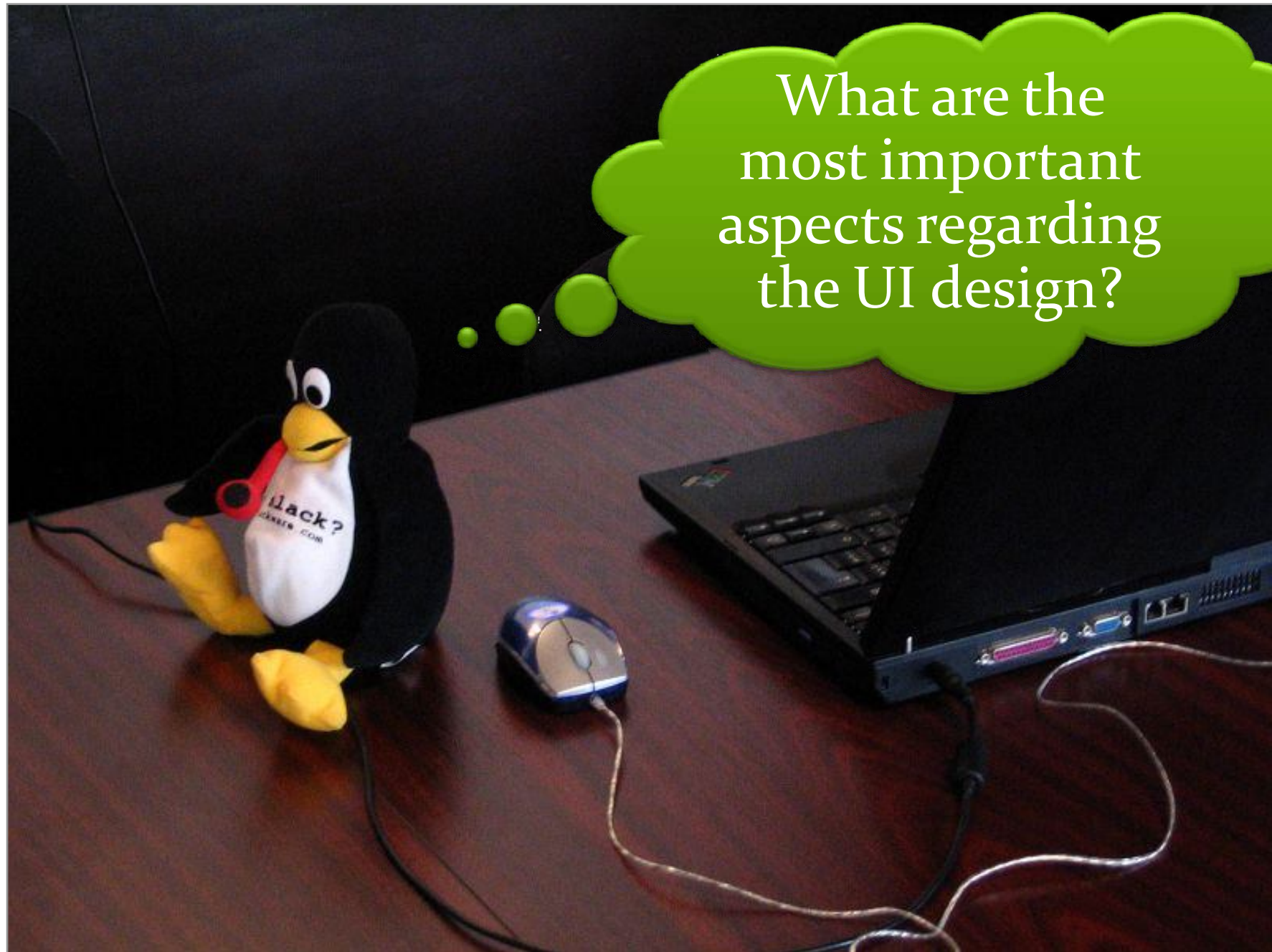
[illegible]



HCI: process



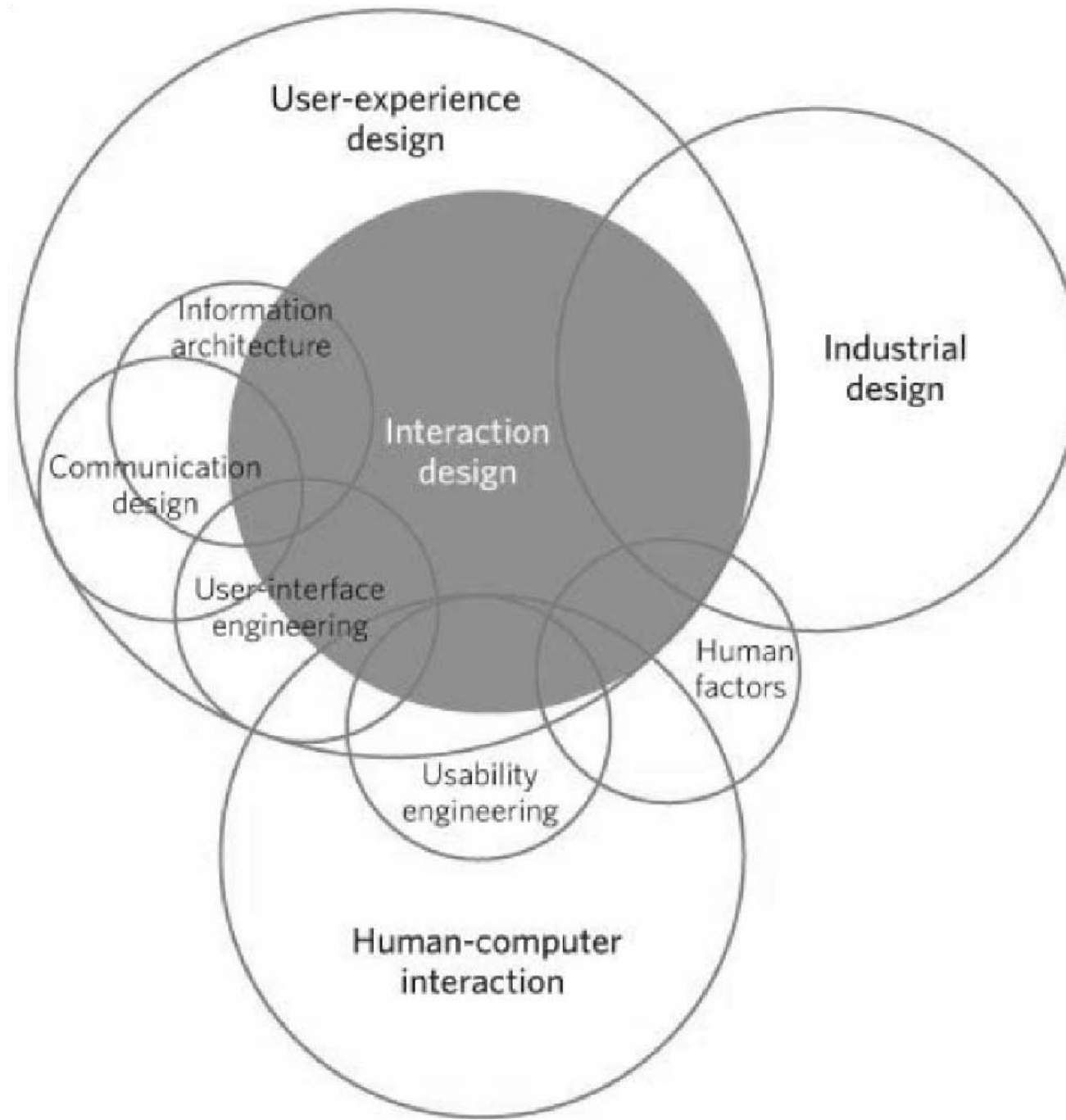
Microsoft, 2008



interaction design

“Designing interactive products to support the way people communicate and interact in their everyday and working lives.”

Sharp, Rogers & Preece, 2007



Related domains (Dan Saffer, 2006)

interaction design

some issues

interaction design

“Digital products are rude.”

Alan Cooper *et al.*, 2007

interaction design

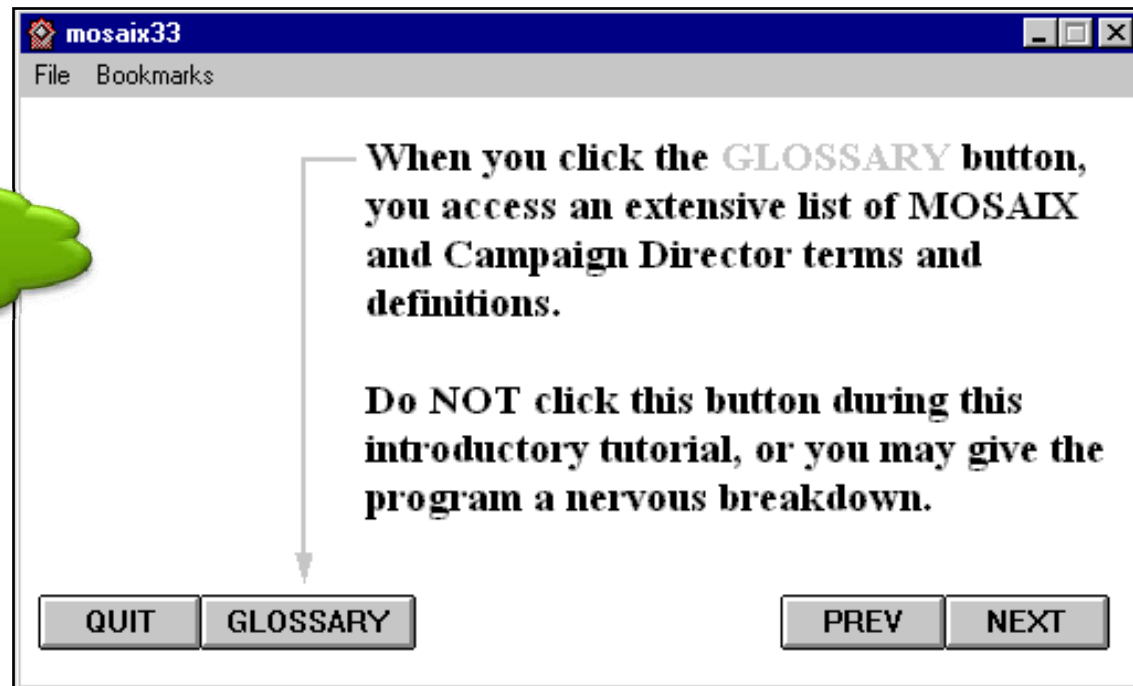
The user is in charge with the application
(developer) mistakes



interaction design

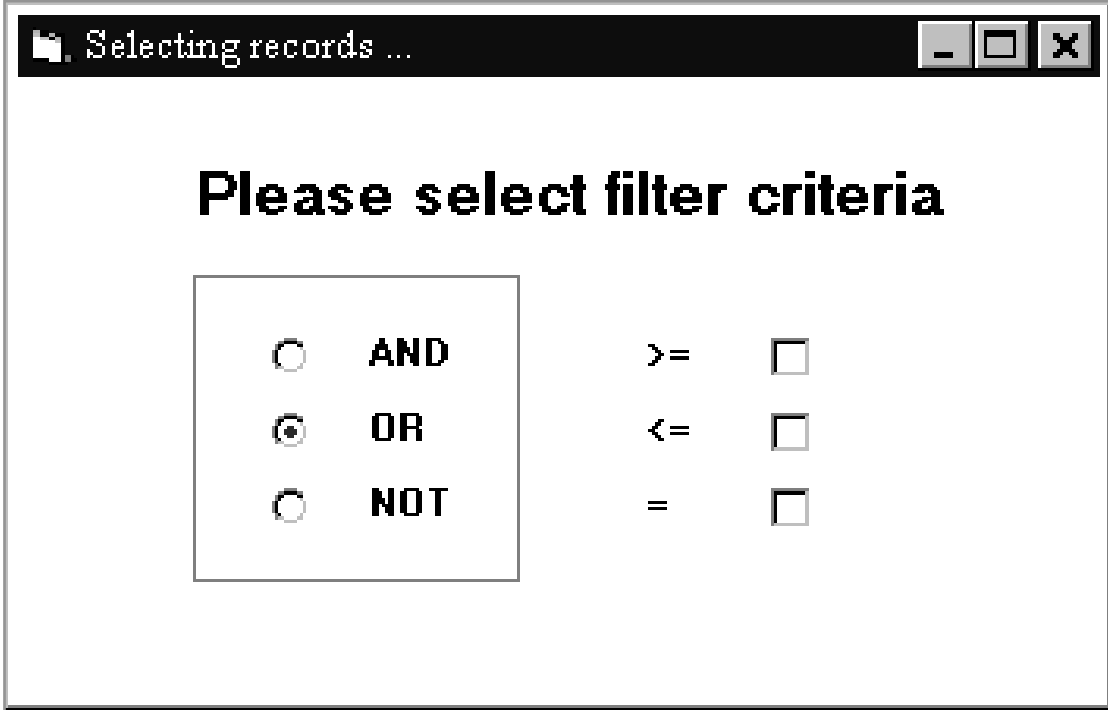
The user is in charge with the application
(developer) mistakes

stupidity?



interaction design

Users must think as computers
...even they don't want to learn the “Boolean” language



A screenshot of a software dialog box titled "Selecting records ...". The dialog box has a standard Windows-style title bar with minimize, maximize, and close buttons. The main content area contains the text "Please select filter criteria". Below this text, there are two columns of options. The first column contains three radio buttons with labels "AND", "OR", and "NOT". The "OR" option is selected, indicated by a filled circle. The second column contains three checkboxes with labels ">=", "<=", and "=".

☐ AND	☐ >=
☒ OR	☐ <=
☐ NOT	☐ =

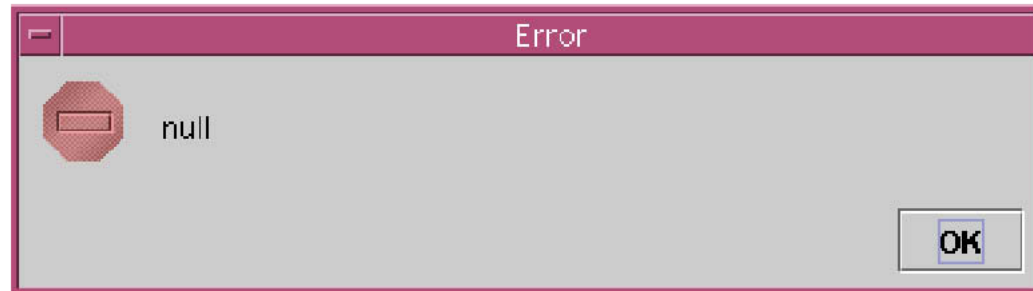
interaction design

Usually, applications do not help people



interaction design

Usually, applications do not help people





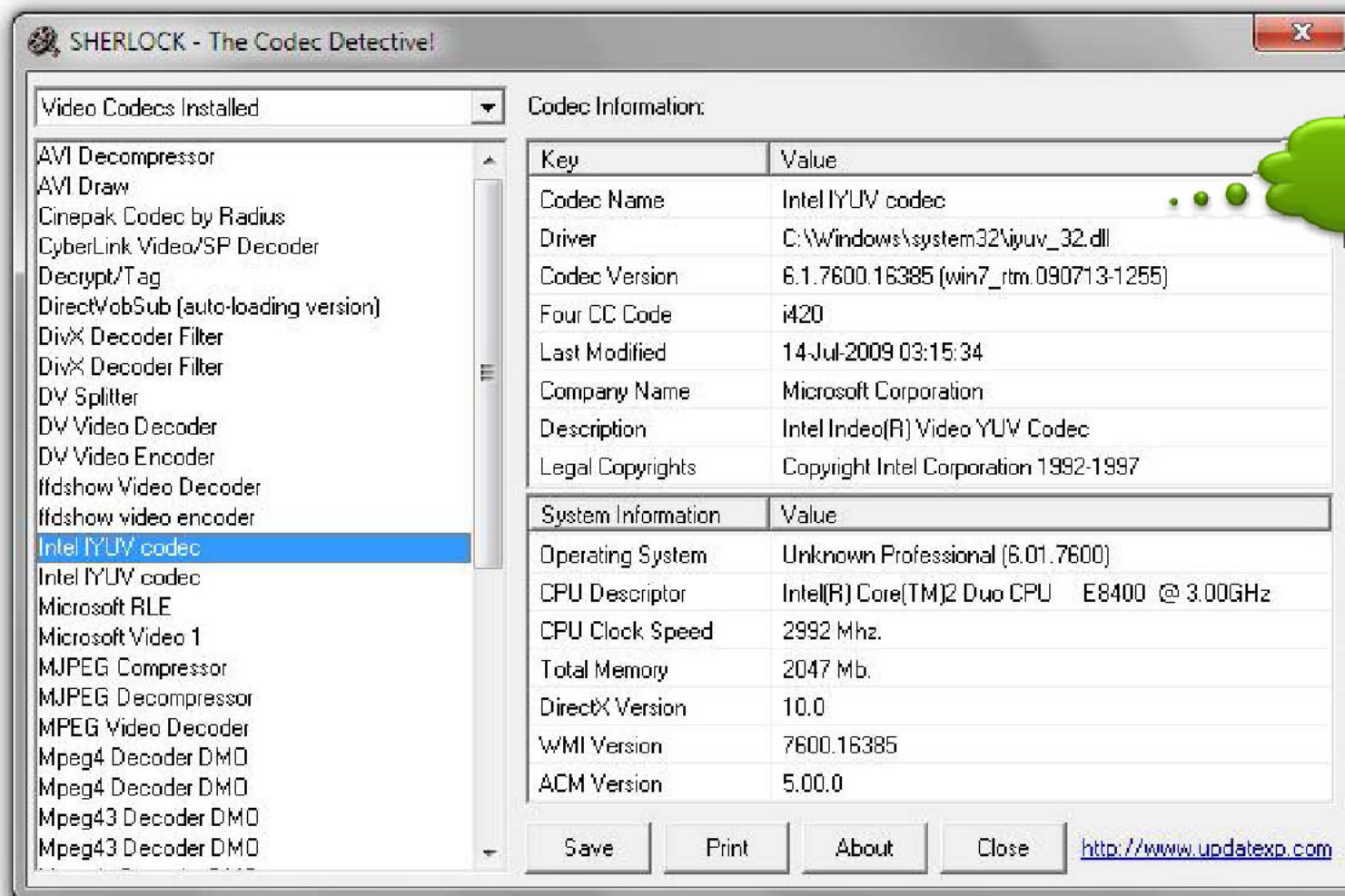
Why digital products are so “nasty”?

interaction design

Ignoring reality aka the target public
(the common user)

we tend exclusively to develop software for the “elite”

interaction design



interaction design

Conflicts of interests:

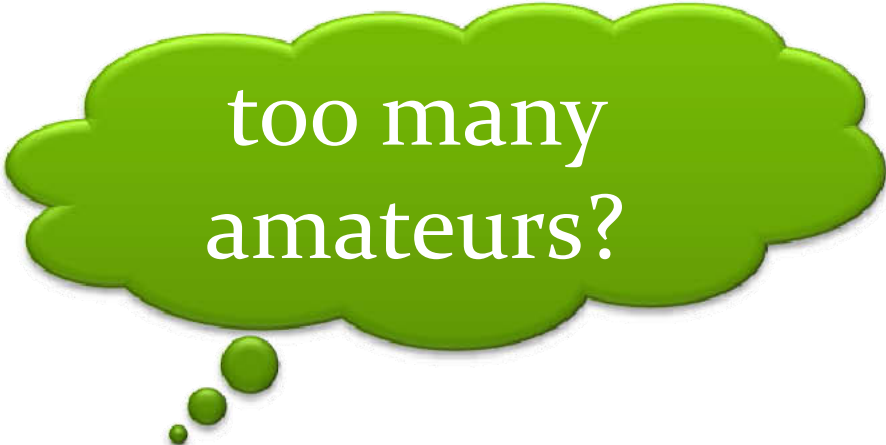
developers
versus
business

interaction design

The lack of a coherent process regarding the developing of the common software

interaction design

The lack of a coherent process regarding the developing of the common software



too many
amateurs?

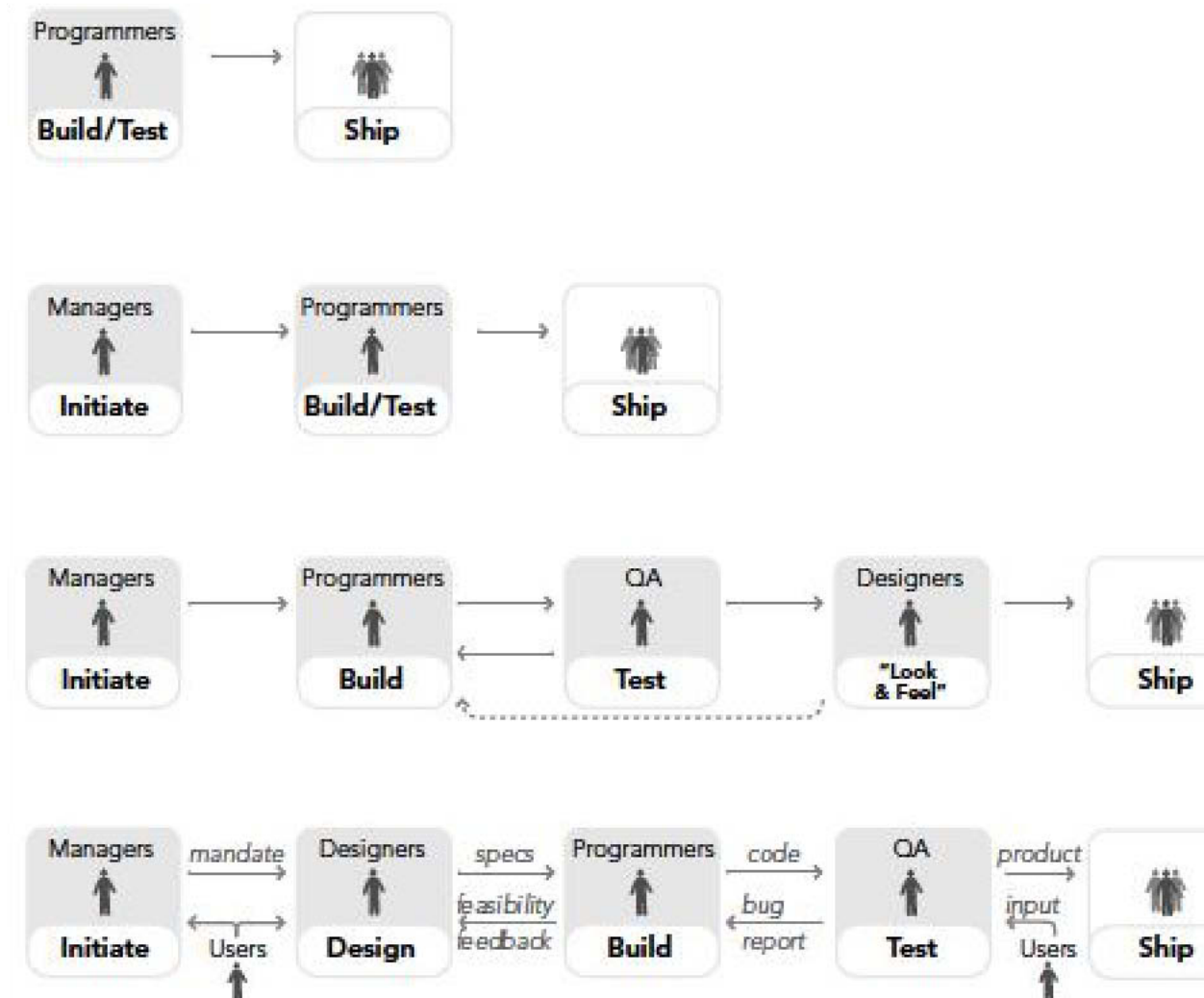
interaction design

The lack of a coherent process regarding the developing of the common software



typical example:
Web
development

interaction design



Alan Cooper et al., 2007

interaction design

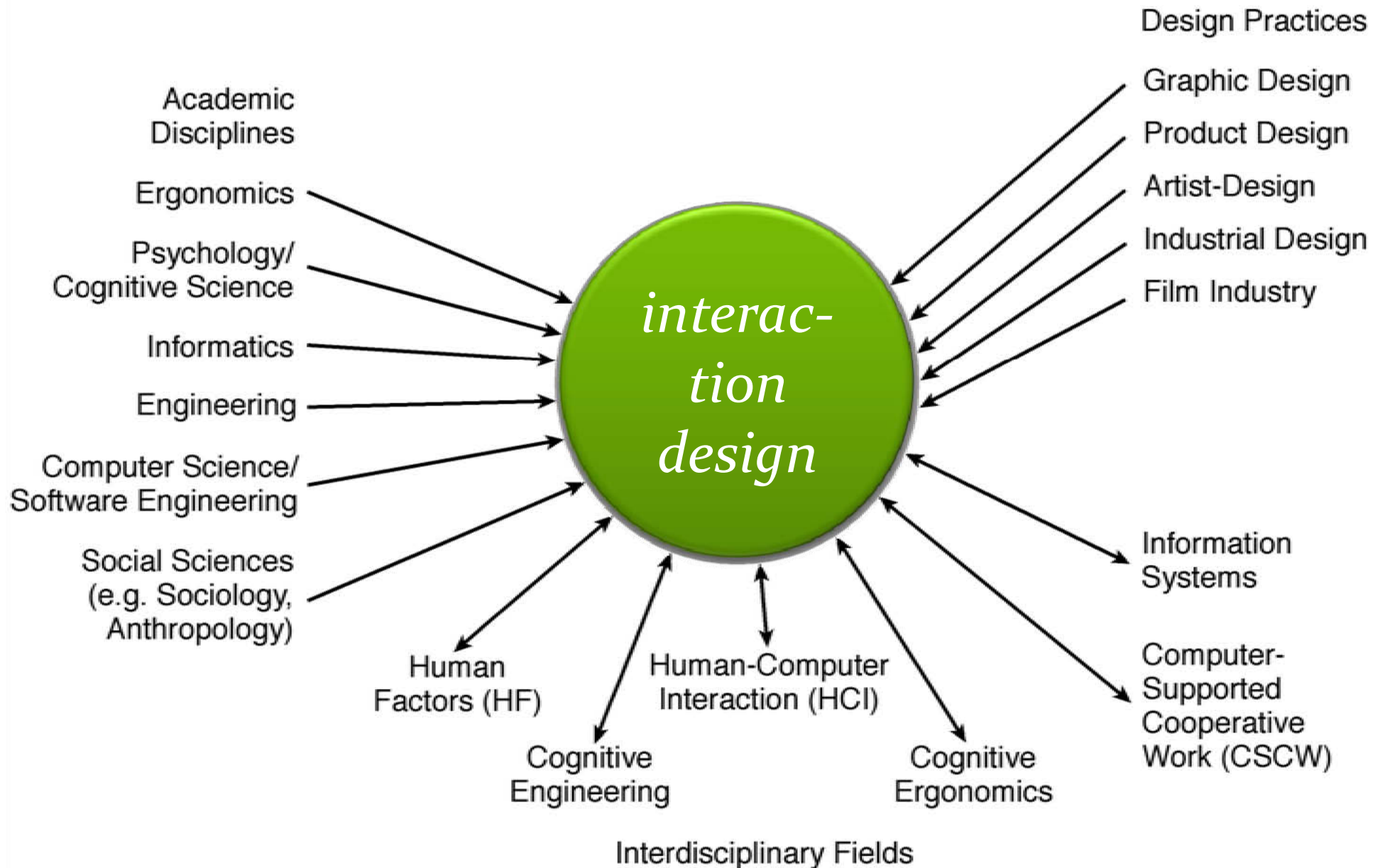
The complexity of software applications is greater than the one of the tangible products

example:
an operating system vs. an airship

interaction design

The process of interaction design is an inter-disciplinary one (Rogers, 2007)





research regarding interaction design (Yvonne Rogers, 2007)

interaction design: questions

What we want to create (develop)?

interaction design: questions

How about the premises?

interaction design: questions

What are the final goals?

interaction design: questions

We'll really obtain what we hope?

interaction design: questions

If yes, how?

interaction design: questions

How about the intrinsic difficulties?

interaction design: questions

Can we (re)use an existing solution?

interaction design

The design process must be performed with respect to the user requirements/goals

user-centered interface design

interaction design

Users (clients) typically speak to you in terms of desired features and solutions, not of needs and problems.

Jenifer Tidwell, 2006

interaction design: questions (II)

How the software will be used?

interaction design: questions (II)

Who will use the developed application?

interaction design: questions (II)

How often software will be used?

interaction design: questions (II)

How long the user will interact
to the application?

interaction design: questions (II)

How about the easiness of using
a given application?

interaction design: questions (II)

Software will be portable?

interaction design: questions (II)

How about the internationalization, localization, accessibility of the application?

interaction design

Important aspect:
understanding the problem to be solved

interaction design

Important aspect:
understanding the problem to be solved

in order to create the desired design (application)

interaction design

Aspects:

UI type(s): desktop, Web, mobile,...

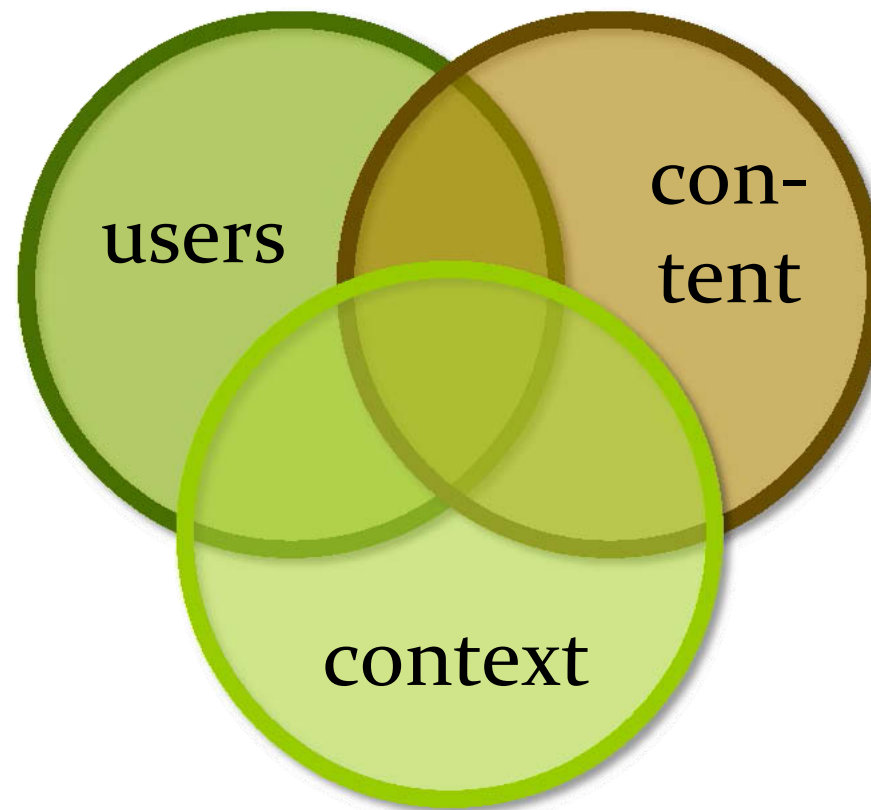
user expected behavior

functionalities

user categories: children, special users, etc.

...

interaction design



interaction design

Study: how a phone call differs regarding the means used to be performed?

using a classical phone unit

versus

via a mobile device: an old and/or last generation phone

versus

using a VoIP software – *e.g.*, Skype

interaction design

We must consider:

(the categories of) involved users
performed activities
the context of using a specific device
the environment

interaction design

The need of a **conceptual model**

describes how the system is perceived by its users

interaction design

The need of a **conceptual model**

software (implementation) model

user (mental) model

environment model

interaction design

The need of a **conceptual model**

“A high level description of how a system is organized and it functions.”

Johnson & Henderson, 2002

interaction design

Important issue:

user model *versus* program model

Alan Cooper *et al.*, 2007

interaction design

Example: the spreadsheet conceptual model

analogy with concrete tables used by accountants

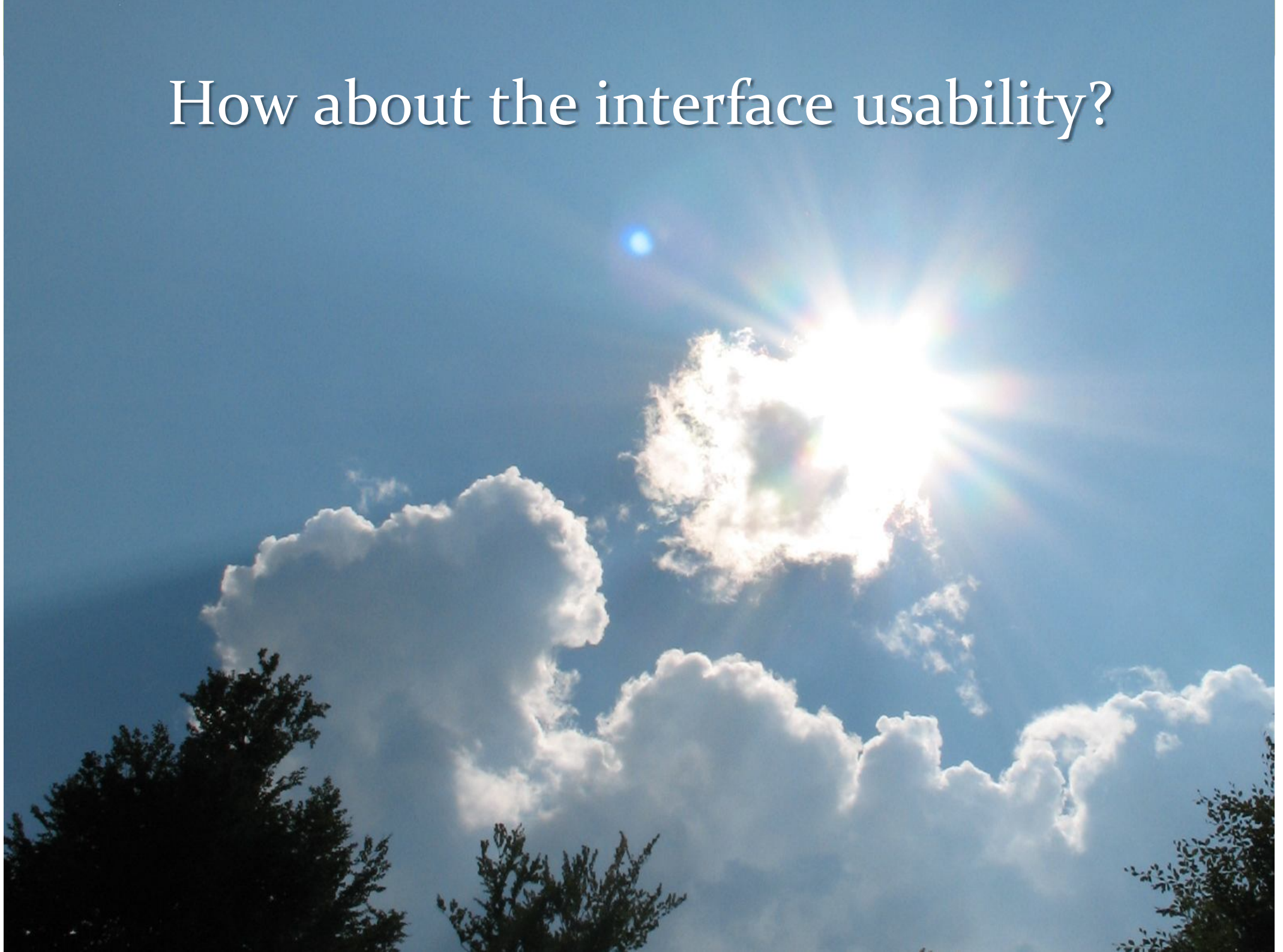
easy to understand

simple interaction

real-time computing of formulas

facilities for further extensions – *e.g.*, URLs as cell values

How about the interface usability?



usability

Usability refers to the “proper” way the users can exploit the functionality of a given system

Jakob Nielsen

www.useit.com/alertbox/20030825.html

usability

Learnability

how easily a person can learn to use a system
(its interface)

user interaction in the context of the *social game computing*

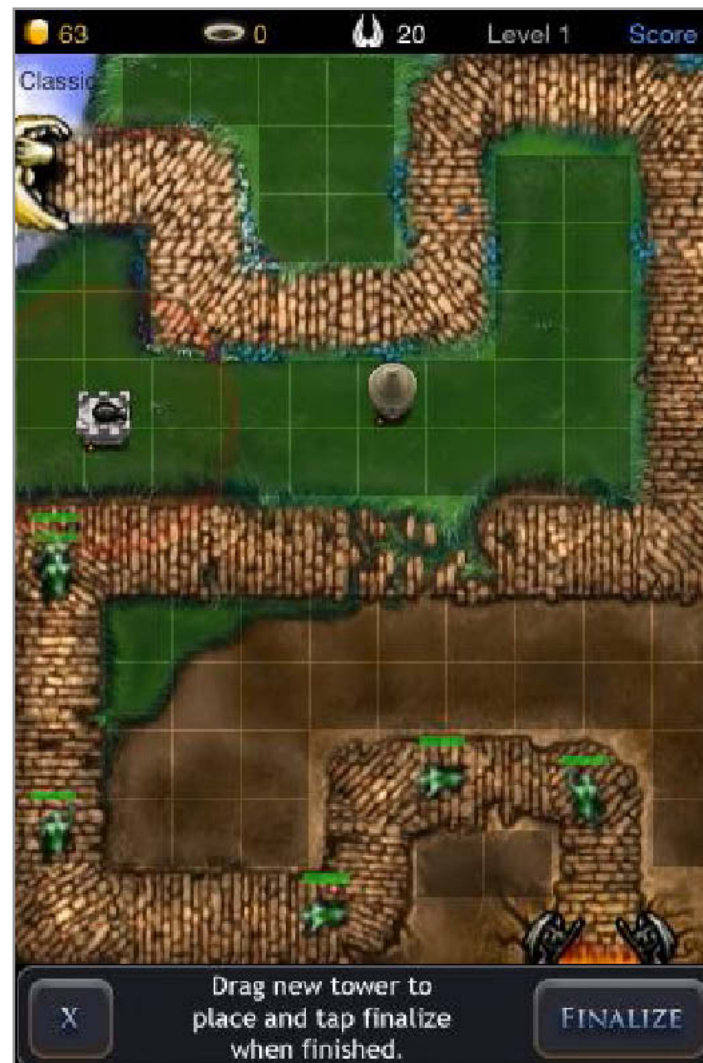


usability

Efficiency

after user learnt the interface,
how can (s)he optimally use it?

usability



usability

Memorability

it is easy for the users to remember
the interaction with the system?

usability



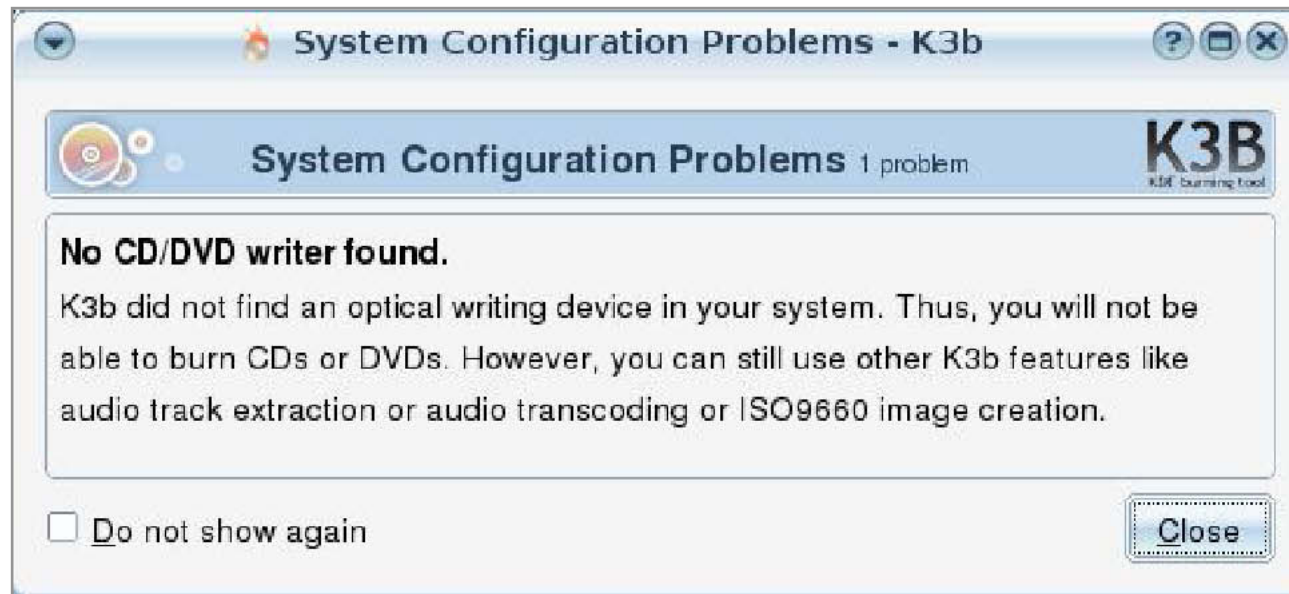
usability

Errors

the number of potential errors must be minimal

the user mistakes must be easily
detected/corrected

usability



usability

Satisfaction

the user likes to use the application/service?

✓ Pass	✓ Pass	Testing	Pending	Pending
				
Video RAM	USB	Drive Hardware	Format Check	Directory Scan

NOKIA N95

ComVu PocketCaster

Birthday party

Broadcasting Live 01:18



Options Exit

READY TO BUY?

 **Add to Shopping Cart**

or

[Sign in](#) to turn on 1-Click ordering.

 A9 Instant Reward Off. [Learn how](#) to reactivate.

remember

“The applications that are easy to use
are designed to be **familiar.**”

Jenifer Tidwell

remember

“Attractive things work better.”

Donald Norman



remember

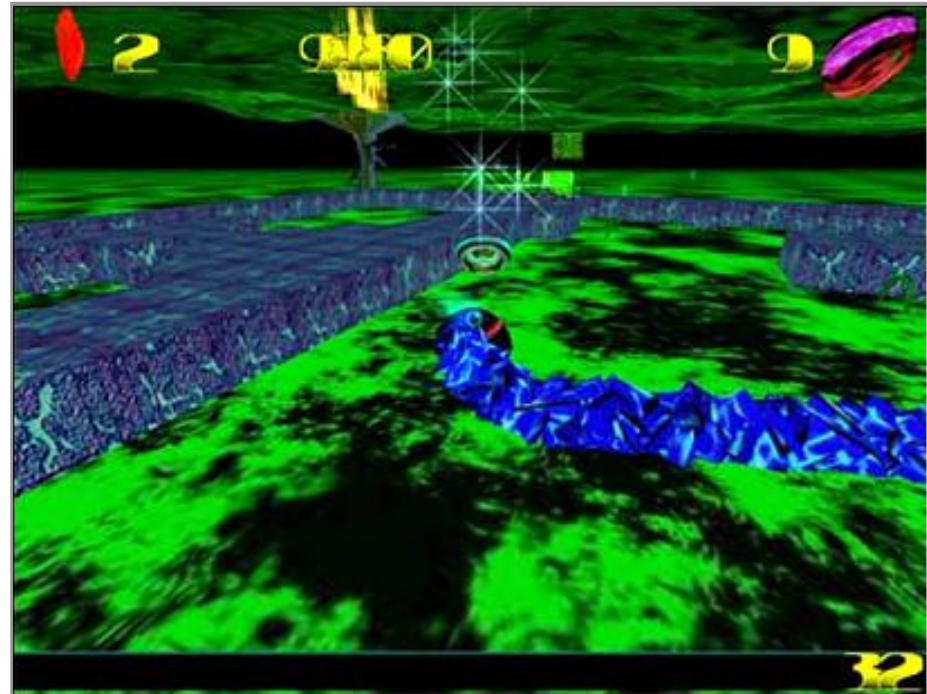
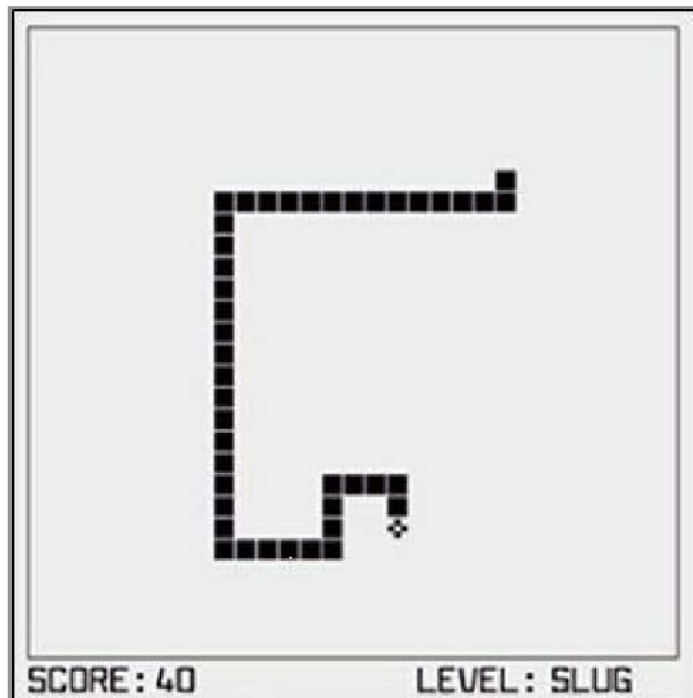
Keep the Simple Simple

“No matter how complex the overall system,
there is no excuse for not keeping
simple tasks simple.”

Jef Raskin

remember

Keep the Simple Simple
...the most engaging interface of a game?



remember

A user-interface is considered to be good if the application is behaving conform to the expectations of its users

remember

A user-interface is considered to be good if the application is behaving conform to the expectations of its users

“The successful games companies are the ones that recognize that their business is **entertainment**, not software.”

Andrew Rollings & Dave Morris, 2004

essential resources

A. Cooper, R. Reimann, D. Cronin,
About Face (3rd Edition), Addison-Wesley, 2007

J. Raskin, *The Humane Interface*, Addison-Wesley, 2000

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interaction, agent, IA, cognetics, markup, art, visual design, KISS, control, user testing, idiom, latency, mode, icon, GUI, Fitt's Law, event, human factors, infographics, RAD, metaphor, accessibility, privacy, skin, affordance, prompt, menu, pointer, risk, augmented reality, UX, animation, web, game, graceful degradation, query, tag, prototyping, ID, evaluation, task, design patterns, gesture, tabs, color, ergonomics, feedback, CLI, script, look & feel, AI, navigation, role, multimodal, audio, RAD, typography, window, voice, mouse, demographics, help, lion, shortcut, direct manipulation, efficiency, paradigm, usability, CADUI, experiment, VR, eye candy, 3D, effect, WYSIWYG, semiotics, i18n, RIA, style, tooltip, iterative, keyboard, pervasive, learnability, hypertext, anticipation, handheld, memory, social, natural language, methodology, ubiquitous computing, persona, dialogue, user, screen, wireframe, PUI, zoom, psychology, recommender system, brand, flow, storyboard, architecture, model, responsiveness, satisfaction, touch, undo, metrics, video, semantic

